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MODULE III

Power Transmission System Calculation of Sag and tension-Insulators
—string efficiency—grading—corona-Characteristics of
transmission lines-Surge Impedance Loading- Series and shunt
compensation.

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Underground cables-ratings- classification- Capacitance —grading-
testing Introduction to EHVAC, HVDC and FACTS: Principle,
classification and advantages/disadvantages



Sag in Overhead lines

- While erecting an overhead line, it is very important that conductors are under safe tension.
- If the conductors are too much stretched between supports in a bid to save conductor material, the stress in the conductor may reach unsafe value and in certain cases the conductor may break due to excessive tension.
- In order to permit safe tension in the conductors, they are not fully stretched but are allowed to have a dip or sag.
- **The difference in level between points of supports and the lowest point on the conductor is called sag**



Fig. shows a conductor suspended between two equilevel supports A and B . The conductor is not fully stretched but is allowed to have a dip. The lowest point on the conductor is O and the sag is S . The following points may be noted :



- (i) When the conductor is suspended between two supports at the same level, it takes the shape of catenary. However, if the sag is very small compared with the span, then sag-span curve is like a parabola.
- (ii) The tension at any point on the conductor acts tangentially. Thus tension T_O at the lowest point O acts horizontally as shown in Fig (ii).
- (iii) The horizontal component of tension is constant throughout the length of the wire.
- (iv) The tension at supports is approximately equal to the horizontal tension acting at any point on the wire. Thus if T is the tension at the support B then $T = T_O$.

Calculation of Sag

- (i) supports are at equal levels and
- (ii) supports are at unequal levels.

(i) When supports are at equal levels. Consider a conductor between two equal level supports *A* and *B* with *O* as the lowest point as shown in Fig. It can be proved that lowest point will be at the mid-span.

Let

l = Length of span

w = Weight per unit length of conductor

T = Tension in the conductor.

Consider a point *P* on the conductor. Taking the lowest point *O* as the origin, let the co-ordinates of point *P* be x and y . Assuming that the curvature is so small that curved length is equal to its horizontal projection (i.e., $OP = x$), the two forces acting on the portion *OP* of the conductor are :

(a) The weight $w x$ of conductor acting at a distance $x/2$ from *O*.

(b) The tension T acting at *O*.

Equating the moments of above two forces about point *O*, we get,

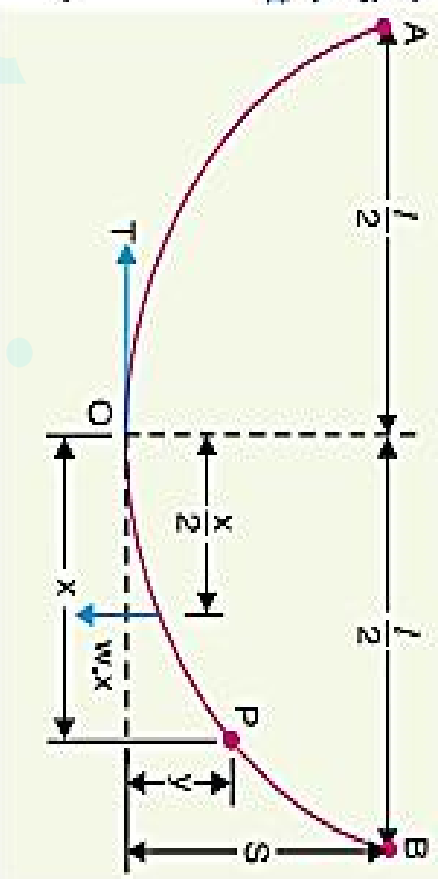
$$T y = w x \times \frac{x}{2}$$

$$\text{or} \quad y = \frac{w x^2}{2 T}$$

The maximum dip (sag) is represented by the value of y at either of the supports *A* and *B*.

At support *A*, $x = l/2$ and $y = S$

$$\therefore \quad S = \frac{w(l/2)^2}{2 T} = \frac{w l^2}{8 T}$$



(ii) **When supports are at unequal levels.** In hilly areas, we generally come across conductors suspended between supports at unequal levels. Fig. 8.25 shows a conductor suspended between two supports *A* and *B* which are at different levels. The lowest point on the conductor is *O*.

Let

l = Span length

h = Difference in levels between two supports

x_1 = Distance of support at lower level (*i.e.*, *A*) from *O*

x_2 = Distance of support at higher level (*i.e.*, *B*) from *O*

T = Tension in the conductor

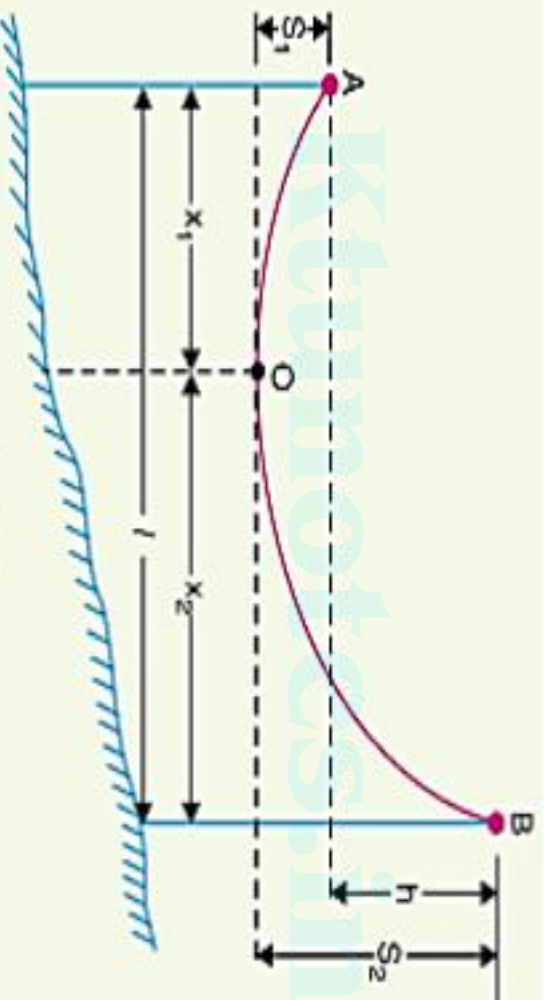


Fig. 8.25

If w is the weight per unit length of the conductor, then,

$$\text{Sag } S_1 = \frac{w x_1^2}{2T}$$

$$\text{and Sag } S_2 = \frac{w x_2^2}{2T}$$

Also

$$x_1 + x_2 = l$$

...(i)

Now

$$S_2 - S_1 = \frac{w}{2T} [x_2^2 - x_1^2] = \frac{w}{2T} (x_2 + x_1)(x_2 - x_1)$$

$\therefore$

$$S_2 - S_1 = \frac{w l}{2T} (x_2 - x_1)$$

$$[\because x_1 + x_2 = l]$$

But

$$S_2 - S_1 = h$$

$\therefore$

$$h = \frac{w l}{2T} (x_2 - x_1)$$

or

$$x_2 - x_1 = \frac{2Th}{wl}$$

...(ii)

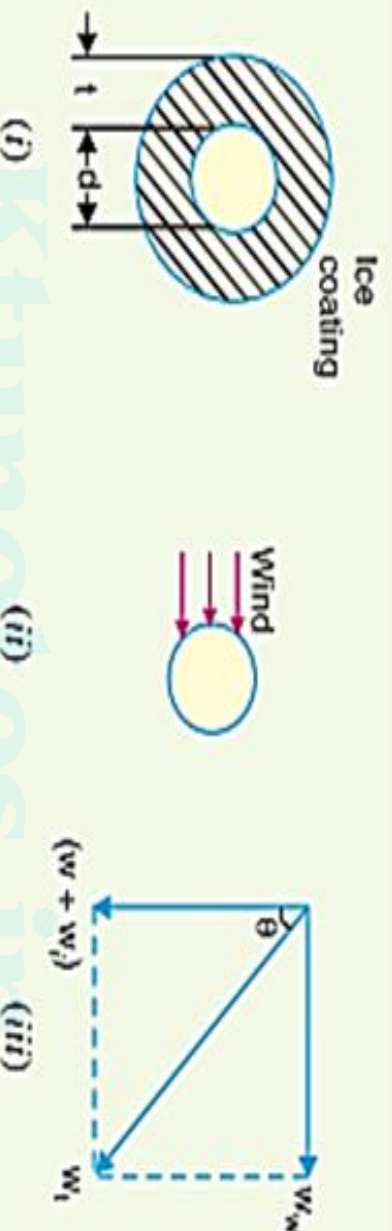
Solving eqns. (i) and (ii), we get,

$$x_1 = \frac{l}{2} - \frac{Th}{wl}$$

$$x_2 = \frac{l}{2} + \frac{Th}{wl}$$

Having found x_1 and x_2 , values of S_1 and S_2 can be easily calculated.

Effect of wind and ice loading. The above formulae for sag are true only in still air and at normal temperature when the conductor is acted by its weight only. However, in actual practice, a conductor may have ice coating and simultaneously subjected to wind pressure. The weight of ice acts vertically downwards *i.e.*, in the same direction as the weight of conductor. The force due to the wind is assumed to act horizontally *i.e.*, at right angle to the projected surface of the conductor. Hence, the total force on the conductor is the vector sum of horizontal and vertical forces as shown in Fig. (iii).



Total weight of conductor per unit length is

$$w_i = \sqrt{(w + w_i)^2 + (w_w)^2}$$

where

w = weight of conductor per unit length

= conductor material density $\times$ volume per unit length

w_i = weight of ice per unit length

= density of ice $\times$ volume of ice per unit length

$$= \text{density of ice} \times \frac{\pi}{4} [(d + 2t)^2 - d^2] \times 1$$

= density of ice $\times \pi t (d + t)^*$

w_w = wind force per unit length

= wind pressure per unit area $\times$ projected area per unit length

When the conductor has wind and ice loading also, the following points may be noted :

- (i) The conductor sets itself in a plane at an angle θ to the vertical where

$$\tan \theta = \frac{w_w}{w + w_i}$$

- (ii) The sag in the conductor is given by :

$$S = \frac{w_i l^2}{2T}$$

Hence S represents the slant sag in a direction making an angle θ to the vertical. *If no specific mention is made in the problem, then slant sag is calculated by using the above formula.*

- (iii) The vertical sag = $S \cos \theta$

$$\text{Working tension, } T = \frac{\text{Ultimate strength}}{\text{Safety factor}}$$

Wt. of conductor/m length, $w = \text{Sp. Gravity} \times \text{Volume of 1 m conductor}$

Tutorial 1

1. A 132 kV transmission line has the following data : Wt. of conductor = 680 kg/km; Length of span = 260 m Ultimate strength = 3100 kg; Safety factor = 2 Calculate the height above ground at which the conductor should be supported. Ground clearance required is 10 metres.

Solution.

Wt. of conductor/metre run, $w = 680/1000 = 0.68 \text{ kg}$

$$\text{Working tension, } T = \frac{\text{Ultimate strength}}{\text{Safety factor}} = \frac{3100}{2} = 1550 \text{ kg}$$

Span length, $l = 260 \text{ m}$

$$\therefore \text{Sag} = \frac{wl^2}{8T} = \frac{0.68 \times (260)^2}{8 \times 1550} = 3.7 \text{ m}$$

$\therefore$ Conductor should be supported at a height of $10 + 3.7 = 13.7 \text{ m}$

Tutorial 1

2. A transmission line has a span of 150 m between level supports. The conductor has a cross-sectional area of 2 cm². The tension in the conductor is 2000 kg. If the specific gravity of the conductor material is 9.9 gm/cm³ and wind pressure is 1.5 kg/m length, calculate the sag. What is the vertical sag?

Solution.

Span length, $l = 150$ m;

Working tension, $T = 2000$ kg

Wind force/m length of conductor, $w_w = 1.5$ kg

Wt. of conductor/m length, $w = \text{Sp. Gravity} \times \text{Volume of 1 m conductor}$
 $= 9.9 \times 2 \times 100 = 1980 \text{ gm} = 1.98 \text{ kg}$

Total weight of 1 m length of conductor is

$$w_t = \sqrt{w^2 + w_w^2} = \sqrt{(1.98)^2 + (1.5)^2} = 2.48 \text{ kg}$$

$$\therefore \text{ Sag, } S = \frac{w_t l^2}{8T} = \frac{2.48 \times (150)^2}{8 \times 2000} = \mathbf{3.48 \text{ m}}$$

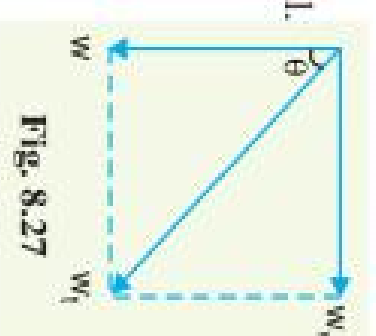
This is the value of slant sag in a direction making an angle θ with the vertical. Referring to Fig. 8.27, the value of θ is given by ;

$$\tan \theta = w_w/w = 1.5/1.98 = 0.76$$

$$\therefore \theta = \tan^{-1} 0.76 = 37.23^\circ$$

$$\therefore \text{ Vertical sag} = S \cos \theta$$

$$= 3.48 \times \cos 37.23^\circ = \mathbf{2.77 \text{ m}}$$



Example 8.19. A transmission line has a span of 200 metres between level supports. The conductor has a cross-sectional area of 1.29 cm^2 , weighs 1170 kg/km and has a breaking stress of 1218 kg/cm^2 . Calculate the sag for a safety factor of 5, allowing a wind pressure of $122 \text{ kg per square metre of projected area}$. What is the vertical sag?

Solution.

Span length,

$$l = 200 \text{ m}$$

Wt. of conductor/m length, $w = 1170/1000 = 1.17 \text{ kg}$

Working tension,

$$*T = 4218 \times 1.29/5 = 1088 \text{ kg}$$

Diameter of conductor,

$$d = \sqrt{\frac{4 \times \text{area}}{\pi}} = \sqrt{\frac{4 \times 1.29}{\pi}} = 1.28 \text{ cm}$$

Wind force/m length,

$$w_w = \text{Pressure} \times \text{projected area in m}^2 \\ = (122) \times (1.28 \times 10^{-2} \times 1) = 1.56 \text{ kg}$$

Total weight of conductor per metre length is

$$w_t = \sqrt{w^2 + w_w^2} = \sqrt{(1.17)^2 + (1.56)^2} = 1.95 \text{ kg}$$

$$\therefore \text{Slant sag, } S = \frac{w_t l^2}{8T} = \frac{1.95 \times (200)^2}{8 \times 1088} = 8.96 \text{ m}$$

The slant sag makes an angle θ with the vertical where value of θ is given by :

$$\theta = \tan^{-1} (w_w/w) = \tan^{-1} (1.56/1.17) = 53.13^\circ$$

$\therefore$

$$\text{Vertical sag} = 8.96 \cos 53.13^\circ = 5.37 \text{ m}$$

Example 8.20. A transmission line has a span of 275 m between level supports. The conductor has an effective diameter of 1.96 cm and weighs 0.865 kg/m. Its ultimate strength is 8060 kg. If the conductor has ice coating of radial thickness 1.27 cm and is subjected to a wind pressure of 3.9 gm/m² of projected area, calculate sag for a safety factor of 2. Weight of 1 c.c. of ice is 0.91 gm.

Solution.

Span length, $l = 275$ m ; Wt. of conductor/m length, $w = 0.865$ kg

Conductor diameter, $d = 1.96$ cm ; Ice coating thickness, $t = 1.27$ cm

Working tension, $T = 8060/2 = 4030$ kg

Volume of ice per metre (i.e., 100 cm) length of conductor

$$= \pi t (d + t) \times 100 \text{ cm}^3$$

$$= \pi \times 1.27 \times (1.96 + 1.27) \times 100 = 1288 \text{ cm}^3$$

Weight of ice per metre length of conductor is

$$w_i = 0.91 \times 1288 = 1172 \text{ gm} = 1.172 \text{ kg}$$

Wind force/m length of conductor is

$$w_w = [\text{Pressure}] \times [(d + 2t) \times 100]$$

$$= [3.9] \times (1.96 + 2 \times 1.27) \times 100 \text{ gm} = 1755 \text{ gm} = 1.755 \text{ kg}$$

Total weight of conductor per metre length of conductor is

$$w_t = \sqrt{(w + w_i)^2 + (w_w)^2}$$

$$= \sqrt{(0.865 + 1.172)^2 + (1.755)^2} = 2.688 \text{ kg}$$

$$\therefore \text{Sag} = \frac{w_t l^2}{2} = \frac{2.688 \times (275)^2}{2} = 6.3 \text{ m}$$

Example 8.21. A transmission line has a span of 214 metres between level supports. The conductors have a cross-sectional area of 3.225 cm^2 . Calculate the factor of safety under the following conditions :

Vertical sag = 2.35 m ;

Wind pressure = 1.5 kg/m run

Breaking stress = 2540 kg/cm^2 ;

Wt. of conductor = 1.125 kg/m run

Solution.

Here,

$$l = 214 \text{ m} ; \quad w = 1.125 \text{ kg} ; \quad w_w = 1.5 \text{ kg}$$

Total weight of one metre length of conductor is

$$w_t = \sqrt{w^2 + w_w^2} = \sqrt{(1.125)^2 + (1.5)^2} = 1.875 \text{ kg}$$

If f is the factor of safety, then,

$$\text{Working tension, } T = \frac{\text{Breaking stress} \times \text{conductor area}}{\text{safety factor}} = 2540 \times 3.225 / f = 8191 / f \text{ kg}$$

$$\text{Slant Sag, } S = \frac{\text{Vertical sag}}{* \cos \theta} = \frac{2.35 \times 1.875}{1.125} = 3.92 \text{ m}$$

Now

$$S = \frac{w_t l^2}{8T}$$

$$\text{or } T = \frac{w_t l^2}{8S}$$

$$\therefore \frac{8191}{f} = \frac{1.875 \times (214)^2}{8 \times 3.92}$$

$$\text{or Safety factor, } f = \frac{8191 \times 8 \times 3.92}{1.875 \times 214^2} = 3$$

Example 8.22. An overhead line has a span of 150 m between level supports. The conductor has a cross-sectional area of 2 cm^2 . The ultimate strength is 5000 kg/cm^2 and safety factor is 5. The specific gravity of the material is 8.9 gm/cc . The wind pressure is 1.5 kg/m . Calculate the height of the conductor above the ground level at which it should be supported if a minimum clearance of 7 m is to be left between the ground and the conductor.

Solution.

Span length, $l = 150 \text{ m}$;

Wind force/m run, $w_w = 1.5 \text{ kg}$

Wt. of conductor/m run,

$$w = \text{conductor area} \times 100 \text{ cm} \times \text{sp. gravity} \\ = 2 \times 100 \times 8.9 = 1780 \text{ gm} = 1.78 \text{ kg}$$

Working tension,

$$T = 5000 \times 2/5 = 2000 \text{ kg}$$

Total weight of one metre length of conductor is

$$w_l = \sqrt{w^2 + w_w^2} = \sqrt{(1.78)^2 + (1.5)^2} = 2.33 \text{ kg}$$

$$\text{Slant sag, } S = \frac{w_l l^2}{8T} = \frac{2.33 \times (150)^2}{8 \times 2000} = 3.28 \text{ m}$$

$$\text{Vertical sag} = S \cos \theta = 3.28 \times w/w_l = 3.28 \times 1.78/2.33 = 2.5 \text{ m}$$

Conductor should be supported at a height of $7 + 2.5 = \mathbf{9.5 \text{ m}}$

Example 8.24. An overhead transmission line conductor having a parabolic configuration weighs 1.925 kg per metre of length. The area of X-section of the conductor is 2.2 cm² and the ultimate strength is 8000 kg/cm². The supports are 600 m apart having 15 m difference of levels. Calculate the sag from the taller of the two supports which must be allowed so that the factor of safety shall be 5. Assume that ice load is 1 kg per metre run and there is no wind pressure.

Solution. Fig. 8.29. shows the conductor suspended between two supports at A and B at different levels with O as the lowest point on the conductor.

Here, $l = 600$ m ; $w_t = 1$ kg ; $h = 15$ m

$$w = 1.925 \text{ kg} ; T = 8000 \times 2.2/5 = 3520 \text{ kg}$$

Total weight of 1 m length of conductor is

$$w_t = w + w_t = 1.925 + 1 = 2.925 \text{ kg}$$

Let the lowest point O of the conductor be at a distance x_1 from the support at lower level (i.e., A) and at a distance x_2 from the support at higher level (i.e., B).

$$\text{Clearly, } x_1 + x_2 = 600 \text{ m} \quad \dots(i)$$

$$\text{Now, } h = S_2 - S_1 = \frac{w_t x_2^2}{2T} - \frac{w_t x_1^2}{2T}$$

$$\text{or } 15 = \frac{w_t}{2T} (x_2 + x_1)(x_2 - x_1)$$

$$\therefore x_2 - x_1 = \frac{2 \times 15 \times 3520}{2.925 \times 600} = 60 \text{ m} \quad \dots(ii)$$

Solving exps. (i) and (ii), we have, $x_1 = 270$ m and $x_2 = 330$ m

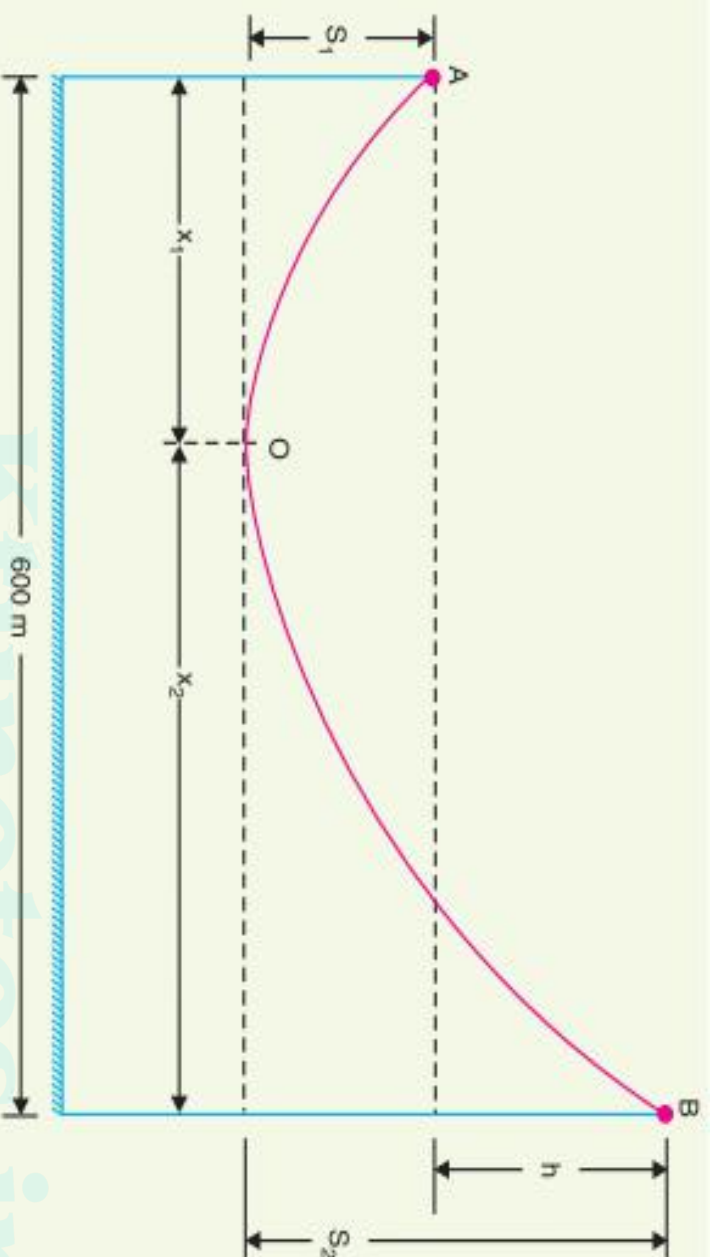


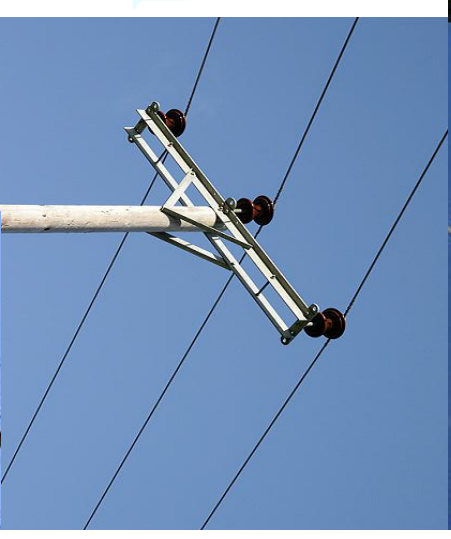
Fig. 8.29

Sag from the taller of the two towers is

$$S_2 = \frac{w_l x_2^2}{2T} = \frac{2 \cdot 925 \times (330)^2}{2 \times 3520} = 45.24 \text{ m}$$

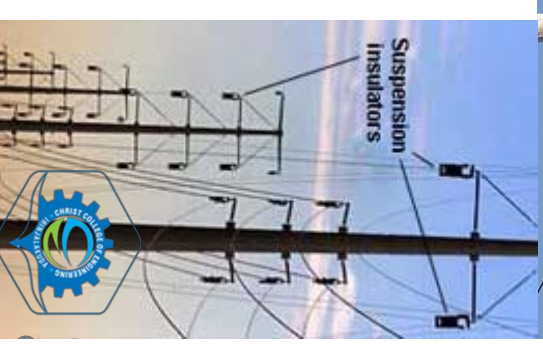
Insulators

- The overhead line conductors should be supported on the poles or towers so that currents from conductors do not flow to earth through supports *i.e.*, line conductors must be properly insulated from supports.



- This is achieved by securing line conductors to supports with the help of *insulators*.

- The insulators provide necessary insulation between line conductors and supports and thus prevent any leakage current from conductors to supports.



Insulator properties

- (i) High mechanical strength in order to withstand conductor load, wind load etc.
- (ii) High electrical resistance of insulator material in order to avoid leakage currents to earth.
- (iii) High relative permittivity of insulator material in order that dielectric strength is high.
- (iv) The insulator material should be non-porous, free from impurities and cracks otherwise the permittivity will be lowered.
- (v) High ratio of puncture strength to flashover.





- The most commonly used material for insulators of overhead line is *porcelain* but glass, steatite and special composition materials are also used to a limited extent.
- Porcelain is produced by firing at a high temperature a mixture of kaolin, feldspar and quartz.

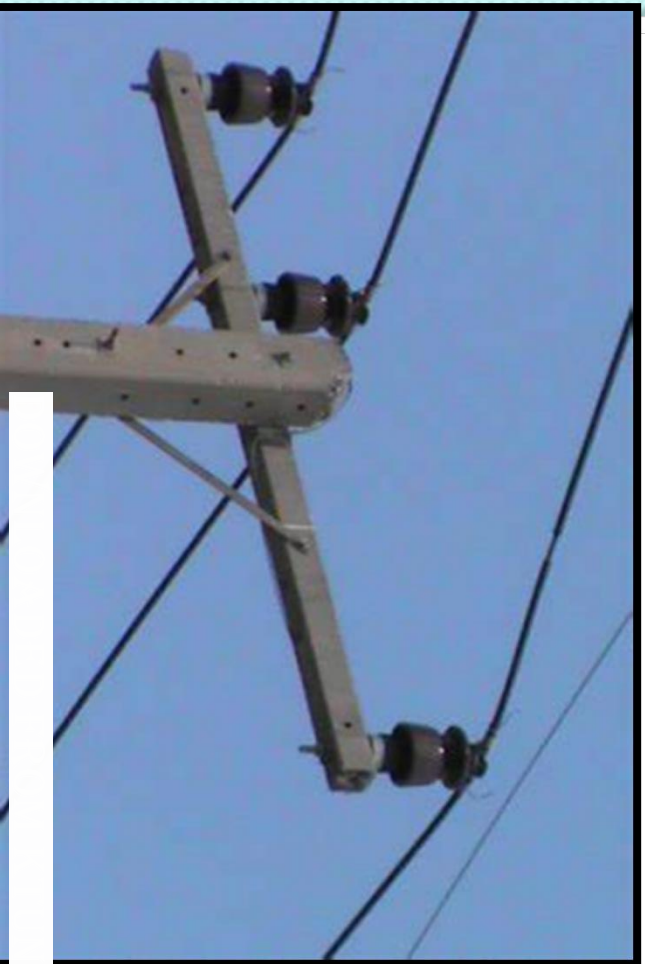
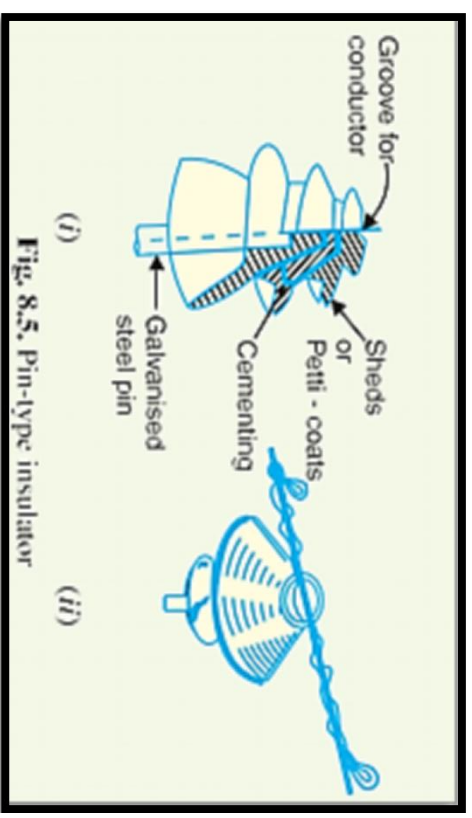
Types of Insulators

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- There are several types of insulators but the most commonly used are
 1. pin type,
 2. suspension type,
 3. strain insulator and
 4. shackle insulator.

Pin type insulators

- The part section of a pin type insulator is shown in Fig.
- As the name suggests, the pin type insulator is secured to the cross-arm on the pole.



- There is a groove on the upper end of the insulator for housing the conductor.
- The conductor passes through this groove and is bound by the annealed wire of the same material as the conductor.
- Pin type insulators are used for transmission and distribution of electric power at voltages upto 33 kV.
- Beyond operating voltage of 33 kV, the pin type insulators become too bulky and hence uneconomical.

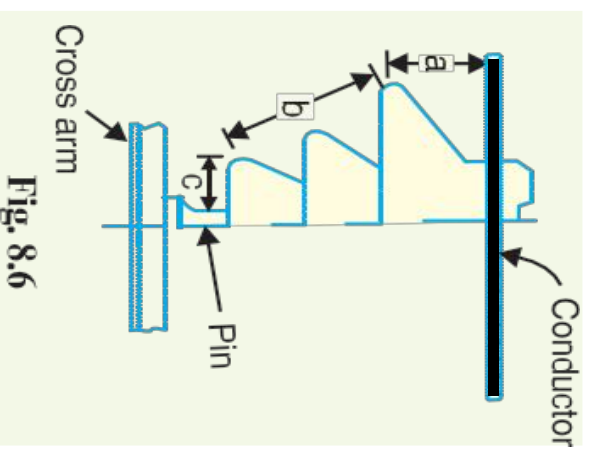


Fig. 8.6

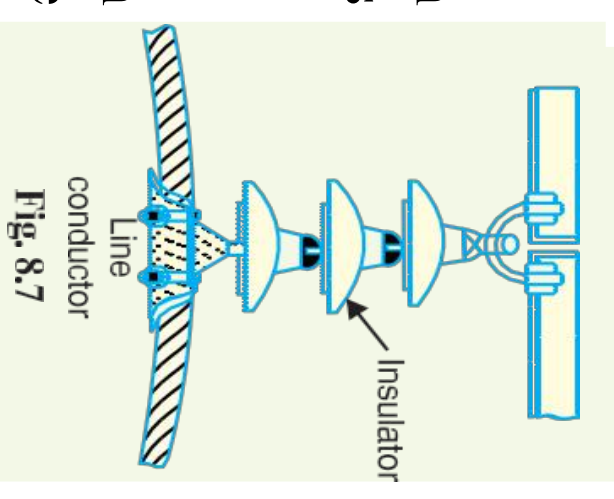


Fig. 8.7

Causes of insulation failure

- Insulators are required to withstand both mechanical and electrical stresses.
- The latter type is primarily due to line voltage and may cause the breakdown of the insulator.
- The electrical break-down of the insulator can occur either by *flash-over* or *puncture*.
- In **flash-over**, an arc occurs between the line conductor and insulator pin (*i.e.*, earth) and the discharge jumps across the air gaps, following shortest distance.

- Fig. shows the arcing distance (*i.e.* $a + b + c$) for the insulator. In case of flash-over, the insulator will continue to act in its proper capacity unless extreme heat produced by the arc destroys the

- In case of **puncture**, the discharge occurs from conductor to pin through the body of the insulator.
-
- When such breakdown is involved, the insulator is permanently destroyed due to excessive heat.
- In practice, sufficient thickness of porcelain is provided in the insulator to avoid puncture by the line voltage.
- The ratio of puncture strength to flash-over voltage is known as safety factor *i.e.*,
- It is desirable that the value of safety factor is high so that flash-over takes place before the insulator gets punctured. For pin type insulators, the value of safety factor is about 10.



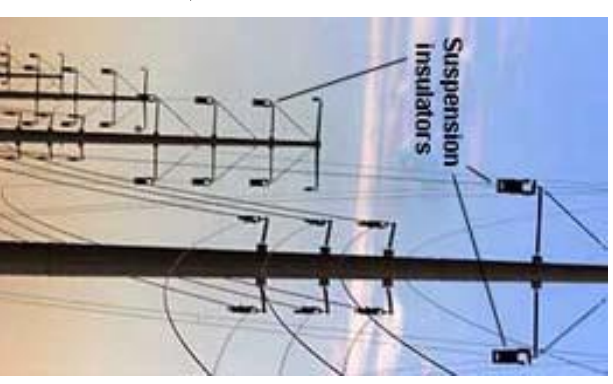
Safety factor of insulator =

$$\frac{\text{Puncture strength}}{\text{Flash - over voltage}}$$



Suspension type insulators

- The cost of pin type insulator increases rapidly as the working voltage is increased.
- Therefore, this type of insulator is not economical beyond 33 kV.
- For high voltages (>33 kV), it is a usual practice to use suspension type insulators shown in Fig.
- They consist of a number of porcelain discs connected in series by metal links in the form of a string.
- The conductor is suspended at the bottom end of this string while the other end of the string is secured to the cross-arm of the tower.
- Each unit or disc is designed for low voltage, say 11 kV. The number of discs in series would obviously depend upon the working voltage.
- For instance, if the working voltage is 66 kV, then six discs in series will be provided on the string.



Advantages

- Suspension type insulators are cheaper than pin type insulators for voltages beyond 33 kV.
- Each unit or disc of suspension type insulator is designed for low voltage, usually 11 kV.
- Depending upon the working voltage, the desired number of discs can be connected in series.
- If any one disc is damaged, the whole string does not become useless because the damaged disc can be replaced by the sound one.
- The suspension type insulators are generally used with steel towers.
- As the conductors run below the earthed cross-arm of the tower, therefore, this arrangement provides partial protection from lightning.



Strain insulators



- When there is a dead end of the line or there is corner or sharp curve, the line is subjected to greater tension. In order to relieve the line of excessive tension, strain insulators are used.

- For low voltage lines ($< 11 \text{ kV}$), shackle insulators are used as strain insulators.

- However, for high voltage transmission lines, strain insulator consists of an assembly of suspension insulators as shown in Fig.

- The discs of strain insulators are used in the vertical plane.

- When the tension in lines is exceedingly high, as at long river spans, two or more strings are used in parallel

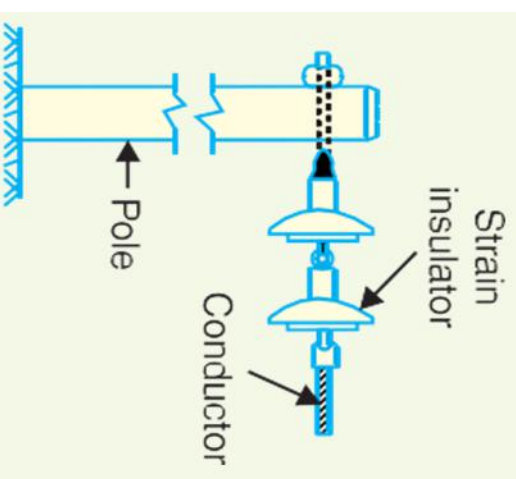
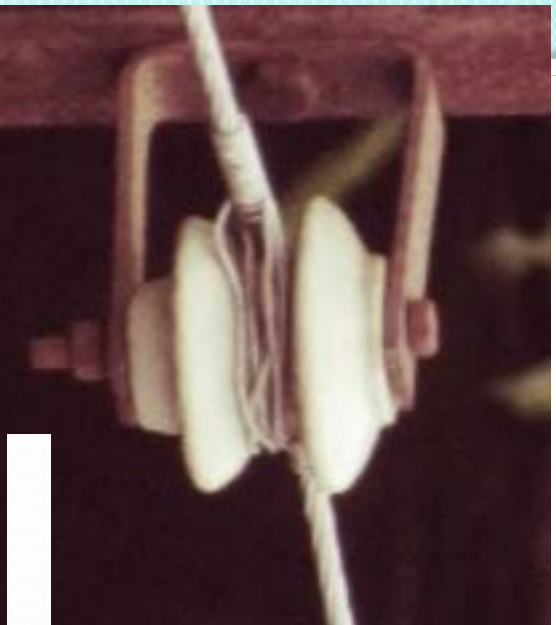


Fig. 8.8. Strain insulator.

Shackle insulators



- In early days, the shackle insulators were used as strain insulators.
- But now a days, they are frequently used for low voltage distribution lines.
- Such insulators can be used either in a horizontal position or in a vertical position.



- They can be directly fixed to the pole with a bolt or to the cross arm.
- Fig. shows a shackle insulator fixed to the pole.

Potential Distribution over Suspension Insulator String

- A string of suspension insulators consists of a number of porcelain discs connected in series through metallic links.
- Fig. (i) shows 3-disc string of suspension insulators. The porcelain portion of each disc is in between two metal links. Therefore, each disc forms a capacitor C as shown in Fig.(ii).
- This is known as ***mutual capacitance or self-capacitance***. If there were mutual capacitance alone, then charging current would have been the same through all the discs and consequently voltage across each unit would have been the same *i.e.*, $V/3$ as shown in Fig.(ii).

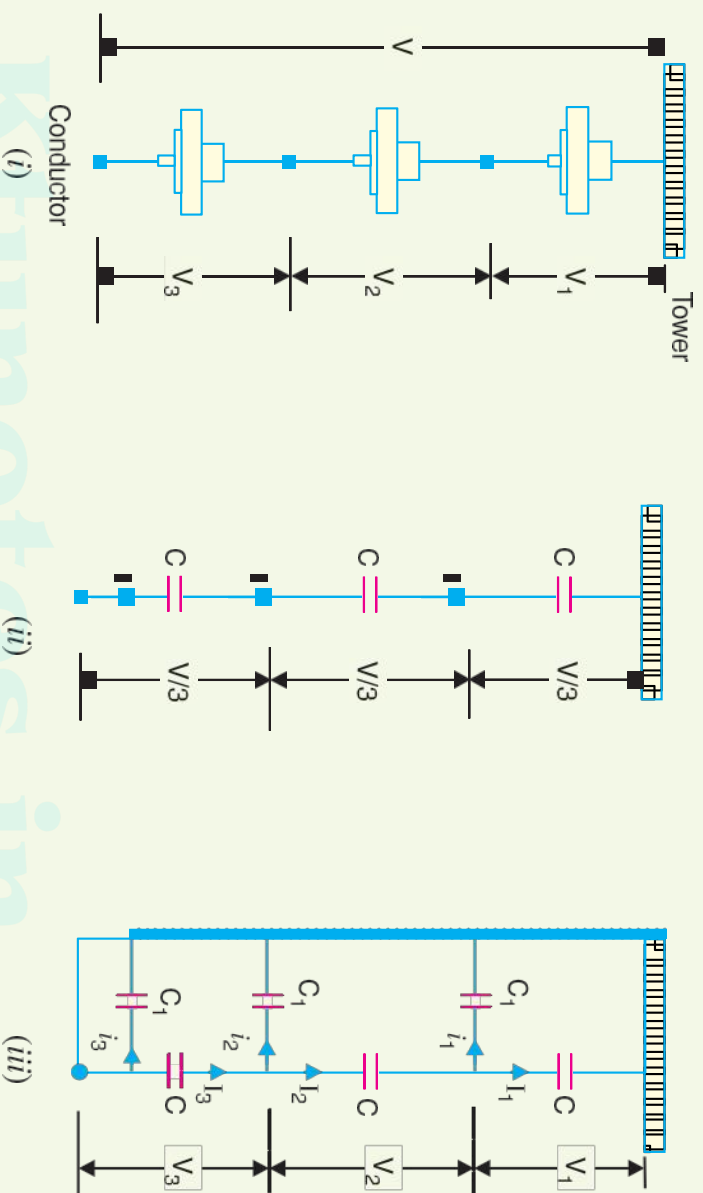


Fig. 8.10

- However, in actual practice, capacitance also exists between metal fitting of each disc and tower or earth. This is known as **shunt capacitance** C_1
- Due to shunt capacitance, charging current is not the same through all the discs of the string (*iii*). Therefore, voltage across each disc will be different.
- Obviously, the disc nearest to the line conductor will have the maximum voltage.
- Thus refer more than V_2 or V_1

- The following points may be noted regarding the potential distribution over a string of suspension insulators :
- The voltage impressed on a string of suspension insulators does not distribute itself uniformly across the individual discs due to the presence of shunt capacitance.
- The disc nearest to the conductor has maximum voltage across it. As we move towards the cross-arm, the voltage across each disc goes on decreasing.
- The unit nearest to the conductor is under maximum electrical stress and is likely to be punctured. Therefore, means must be provided to equalise the potential across each unit.
- If the voltage impressed across the string were d.c., then voltage across each unit would be the same. It is because insulator capacitances are ineffective for d.c.



String Efficiency

- As stated above, the voltage applied across the string of suspension insulators is not uniformly distributed across various units or discs.
- The disc nearest to the conductor has much higher potential than the other discs.
- This unequal potential distribution is undesirable and is usually expressed in terms of string efficiency.
- *The ratio of voltage across the whole string to the product of number of discs and the voltage across the disc nearest to the conductor is known as string efficiency i.e.,*

$$\text{String efficiency} = \frac{\text{Voltage across the string}}{n \times \text{Voltage across disc nearest to conductor}}$$

n = number of discs in the string.

- String efficiency is an important consideration since it decides the potential distribution along the string.
- The greater the string efficiency, the more uniform is the voltage distribution.
- Thus 100% string efficiency is an ideal case for which the voltage across each disc will be exactly the same.
- Although it is impossible to achieve 100% string efficiency, yet efforts should be made to improve it as close to this value as possible

Mathematical expression

- Fig shows the equivalent circuit for a 3-disc string.
- Let us suppose that self capacitance of each disc is C .
- Let us further assume that shunt capacitance C_1 is some fraction K of self- capacitance *i.e.*,
- $C_1 = KC$.
- Starting from the cross-arm or tower, the voltage across each unit is V_1, V_2 and V_3 respectively as shown.

Applying Kirchhoff's current law to node A, we get,

$$I_2 = I_1 + i_1$$

or

$$V_2 \omega C^* = V_1 \omega C + V_1 \omega C_1$$

or

$$V_2 \omega C = V_1 \omega C + V_1 \omega K C$$

$\therefore$

$$V_2 = V_1 (1 + K)$$

...(i)

Applying Kirchhoff's current law to node B, we get,

$$I_3 = I_2 + i_2$$

or

$$V_3 \omega C = V_2 \omega C + (V_1 + V_2) \omega C_1 +$$

or

$$V_3 \omega C = V_2 \omega C + (V_1 + V_2) \omega K C$$

or

$$V_3 = V_2 + (V_1 + V_2) K$$

$$= K V_1 + V_2 (1 + K)$$

$$= K V_1 + V_1 (1 + K)^2$$

$$= V_1 [K + (1 + K)^2]$$

$\therefore$

$$V_3 = V_1 [1 + 3K + K^2]$$

Voltage between conductor and earth (i.e., tower) is

$$V = V_1 + V_2 + V_3$$

$$= V_1 + V_1 (1 + K) + V_1 (1 + 3K + K^2)$$

$$= V_1 (3 + 4K + K^2)$$

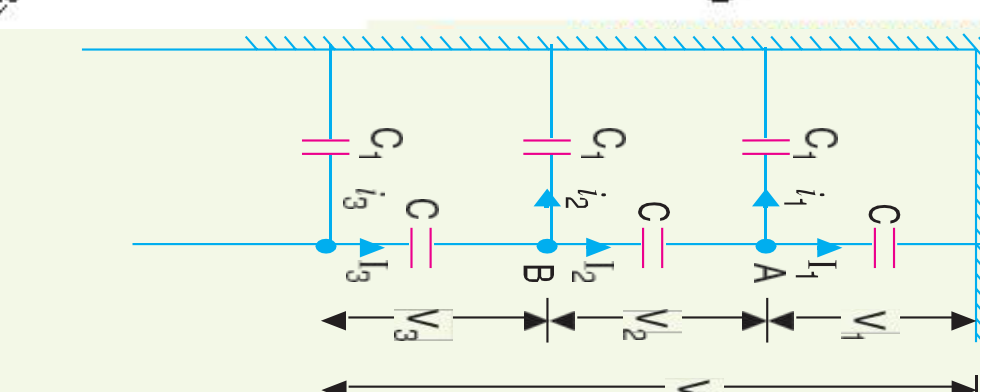
$$V = V_1 (1 + K) (3 + K)$$

$\therefore$

From expressions (i), (ii) and (iii), we get,

$$\frac{V_1}{1} = \frac{V_2}{1 + K} = \frac{V_3}{1 + 3K + K^2} = \frac{V}{(1 + K)(3 + K)}$$

$\therefore$ Voltage across top unit $V_1 = \frac{V}{(1 + K)(3 + K)}$



Voltage across second unit from top, $V_2 = V_1 (1 + K)$

Voltage across third unit from top, $V_3 = V_1 (1 + 3K + K^2)$

$$\begin{aligned}\% \text{age String efficiency} &= \frac{\text{Voltage across string}}{n \times \text{Voltage across disc nearest to conductor}} \times 100 \\ &= \frac{V}{3 \times V_3} \times 100\end{aligned}$$

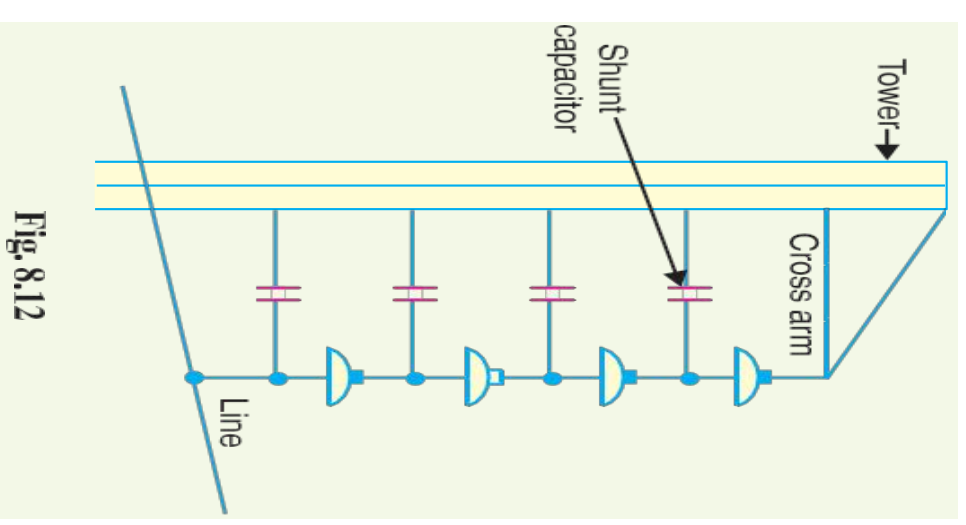
- The following points may be noted from the above mathematical analysis :
- If $K = 0.2$ (Say), then from exp. (iv),
- we get, $V_2 = 1.2 V_1$ and $V_3 = 1.64 V_1$.
- This clearly shows that disc nearest to the conductor has maximum voltage across it; the voltage across other discs decreasing progressively as the cross-arm is approached.
- The greater the value of K ($= C1/C$), the more non-uniform is the potential across the discs and lesser is the string efficiency.
- The inequality in voltage distribution increases with the increase of number of discs in the string. Therefore, shorter string has more efficiency than the larger one.

Methods of Improving String Efficiency

- It has been seen above that potential distribution in a string of suspension insulators is not uniform.
- The maximum voltage appears across the insulator nearest to the line conductor and decreases progressively as the cross-arm is approached.
- If the insulation of the highest stressed insulator (*i.e.* nearest to conductor) breaks down or flash over takes place, the breakdown of other units will take place in succession.
- This necessitates to equalise the potential across the various units of the string *i.e.* to improve the string efficiency.
- The various methods for this purpose are :

1. *By using longer cross-arms*

- The value of string efficiency depends upon the value of K i.e., ratio of shunt capacitance to mutual capacitance.
- The lesser the value of K , the greater is the string efficiency and more uniform is the voltage distribution.
- The value of K can be decreased by reducing the shunt capacitance.
- In order to reduce shunt capacitance, the distance of conductor from tower must be increased i.e., longer cross-arms should be used.
- However, limitations of cost and strength of tower do not allow the use of very long cross-arms.
- In practice, $K = 0.1$ is the limit that can be achieved by this method.



2. *By grading the insulators*

- In this method, insulators of different dimensions are so chosen that each has a different capacitance.
- The insulators are capacitance graded *i.e.* they are assembled in the string in such a way that the top unit has the minimum capacitance, increasing progressively as the bottom unit (*i.e.*, nearest to conductor) is reached.
- Since voltage is inversely proportional to capacitance, this method tends to equalise the potential distribution across the units in the string.
- This method has the disadvantage that a large number of different-sized insulators are required.
- However, good results can be obtained by using standard insulators for most of the string and larger units for that near to the line conductor.

3. *By using a guard ring*

- The potential across each unit in a string can be equalised by using a guard ring which is a metal ring electrically connected to the conductor and surrounding the bottom insulator as shown in the Fig.
- The guard ring introduces capacitance between metal fittings and the line conductor.
- The guard ring is contoured in such a way that shunt capacitance currents i_1 , i_2 etc. are equal to metal fitting line capacitance currents i'_1 , i'_2 etc.
- The result is that same charging current I flows through each unit of string.
- Consequently, there will be uniform potential distribution across the units

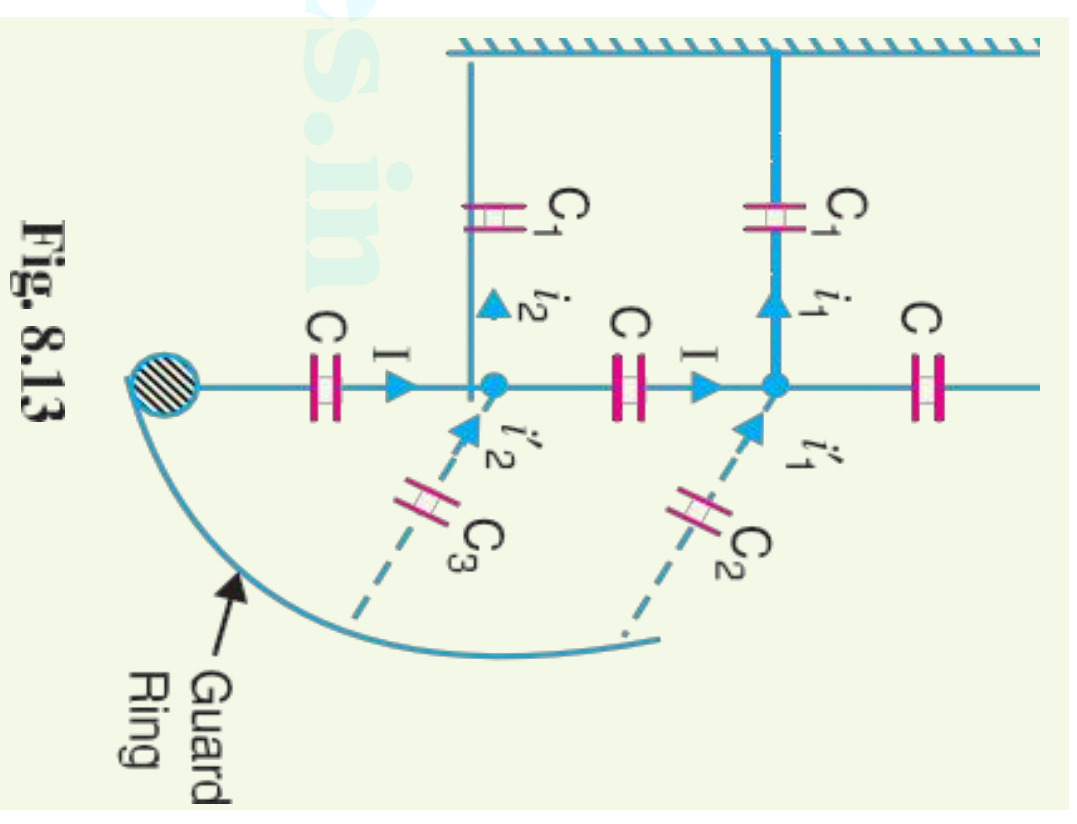


Fig. 8.13

Important Points

- While solving problems relating to string efficiency, the following points must be kept in mind:
- The maximum voltage appears across the disc nearest to the conductor (*i.e.*, line conductor).
- The voltage across the string is equal to phase voltage *i.e.*,
- Voltage across string = Voltage between line and earth = Phase Voltage
- Line Voltage = $\sqrt{3} \times \text{Voltage across string}$

Tutorial2

- **Example** *In a 33 kV overhead line, there are three units in the string of insulators. If the capacitance between each insulator pin and earth is 11% of self-capacitance of each insulator, find (i) the distribution of voltage over 3 insulators and (ii) string efficiency.*

Solution. Fig. 8.14 shows the equivalent circuit of string insulators. Let V_1 , V_2 and V_3 be the voltage across top, middle and bottom unit respectively. If C is the self-capacitance of each unit, then KC will be the shunt capacitance.

$$K = \frac{\text{Shunt Capacitance}}{\text{Self-capacitance}} = 0.11$$

$$\text{Voltage across string, } V = 33/\sqrt{3} = 19.05 \text{ kV}$$

At Junction A

$$I_2 = I_1 + i_1$$

$$V_2 \omega C = V_1 \omega C + V_1 K \omega C$$

$$V_2 = V_1 (1 + K) = V_1 (1 + 0.11)$$

$$V_2 = 1.11 V_1$$

... (i)

At Junction B

$$I_3 = I_2 + i_2$$

$$V_3 \omega C = V_2 \omega C + (V_1 + V_2) K \omega C$$

$$V_3 = V_2 + (V_1 + V_2) K$$

$$= 1.11 V_1 + (V_1 + 1.11 V_1) 0.11$$

$$V_3 = 1.342 V_1$$

(ii) Voltage across the whole string is

$$V = V_1 + V_2 + V_3 = V_1 + 1.11 V_1 + 1.342 V_1 = 3.452 V_1$$

$$\text{or } 19.05 = 3.452 V_1$$

$$\therefore \text{Voltage across top unit, } V_1 = 19.05/3.452 = \mathbf{5.52 \text{ kV}}$$

$$\text{Voltage across middle unit, } V_2 = 1.11 V_1 = 1.11 \times 5.52 = \mathbf{6.13 \text{ kV}}$$

$$\text{Voltage across bottom unit, } V_3 = 1.342 V_1 = 1.342 \times 5.52 = \mathbf{7.4 \text{ kV}}$$

$$\text{(iii) String efficiency} = \frac{\text{Voltage across string}}{\text{Voltage across unit}} \times 100 = \frac{19.05}{7.4} \times 100 = \mathbf{85.8\%}$$

$$\text{No. of insulators} \times V_3 \quad 3 \times 7.4$$

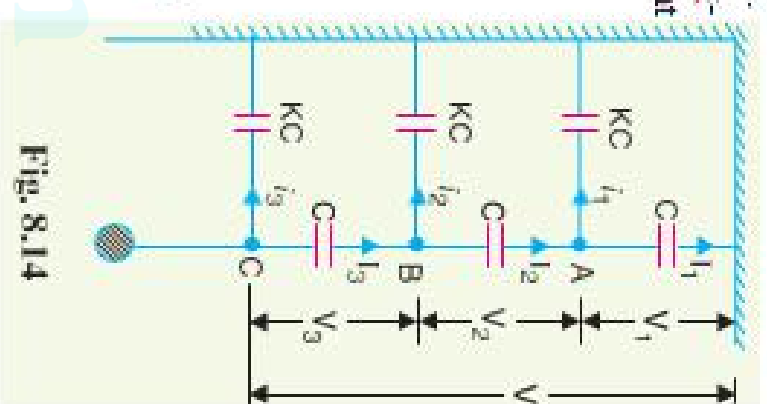


Fig. 8.14

Example 8.2. A 3-phase transmission line is being supported by three disc insulators. The potentials across top unit (i.e., near to the tower) and middle unit are 8 kV and 11 kV respectively. Calculate (i) the ratio of capacitance between pin and earth to the self-capacitance of each unit (ii) the line voltage and (iii) string efficiency.

Solution. The equivalent circuit of string insulators is the same as shown in Fig. 8.14. It is given that $V_1 = 8 \text{ kV}$ and $V_2 = 11 \text{ kV}$.

(i) Let K be the ratio of capacitance between pin and earth to self capacitance. If C farad is the self capacitance of each unit, then capacitance between pin and earth = KC .

Applying Kirchoff's current law to Junction A,

$$I_2 = I_1 + i_1$$

$$\text{or } V_2 \omega C = V_1 \omega C + V_1 K \omega C$$

$$\text{or } V_2 = V_1 (1 + K)$$

$$\therefore K = \frac{V_2 - V_1}{V_1} = \frac{11 - 8}{8} = 0.375$$

(ii) Applying Kirchoff's current law to Junction B,

$$I_3 = I_2 + i_2$$

$$\text{or } V_3 \omega C = V_2 \omega C + (V_1 + V_2) K \omega C$$

$$\text{or } V_3 = V_2 + (V_1 + V_2) K = 11 + (8 + 11) \times 0.375 = 18.12 \text{ kV}$$

$$\text{Voltage between line and earth} = V_1 + V_2 + V_3 = 8 + 11 + 18.12 = 37.12 \text{ kV}$$

$$\therefore \text{Line Voltage} = \sqrt{3} \times 37.12 = 64.28 \text{ kV}$$

$$(iii) \text{ String efficiency} = \frac{\text{Voltage across string}}{\text{No. of insulators} \times V_3} \times 100 = \frac{37.12}{3 \times 18.12} \times 100 = 68.28\%$$

Example 8.3. Each line of a 3-phase system is suspended by a string of 3 similar insulators. If the voltage across the line unit is 17.5 kV, calculate the line to neutral voltage. Assume that the shunt capacitance between each insulator and earth is 1/8th of the capacitance of the insulator itself. Also find the string efficiency.

Solution. Fig. 8.15 shows the equivalent circuit of string insulators. If C is the self capacitance of each unit, then KC will be the shunt capacitance where $K = 1/8 = 0.125$.

Voltage across line unit, $V_3 = 17.5$ kV

At Junction A

$$I_2 = I_1 + i_1$$

$$V_2 \omega C = V_1 \omega C + V_1 K \omega C$$

$$\text{or } V_2 = V_1 (1 + K) = V_1 (1 + 0.125)$$

$$\therefore V_2 = 1.125 V_1$$

At Junction B

$$I_3 = I_2 + i_2$$

$$\text{or } V_3 \omega C = V_2 \omega C + (V_1 + V_2) K \omega C$$

$$\text{or } V_3 = V_2 + (V_1 + V_2) K$$

$$= 1.125 V_1 + (V_1 + 1.125 V_1) \times 0.125$$

$$\therefore V_3 = 1.39 V_1$$

$$\begin{aligned} \text{Voltage across top unit, } V_1 &= V_3 / 1.39 = 17.5 / 1.39 \\ &= 12.59 \text{ kV} \end{aligned}$$

$$\text{Voltage across middle unit, } V_2 = 1.125 V_1 = 1.125 \times 12.59 = 14.16 \text{ kV}$$

$\therefore$ Voltage between line and earth (i.e., line to neutral)

$$= V_1 + V_2 + V_3 = 12.59 + 14.16 + 17.5 = \mathbf{44.25 \text{ kV}}$$

$$\text{String efficiency} = \frac{44.25}{44} \times 100 = \mathbf{84.78\%}$$

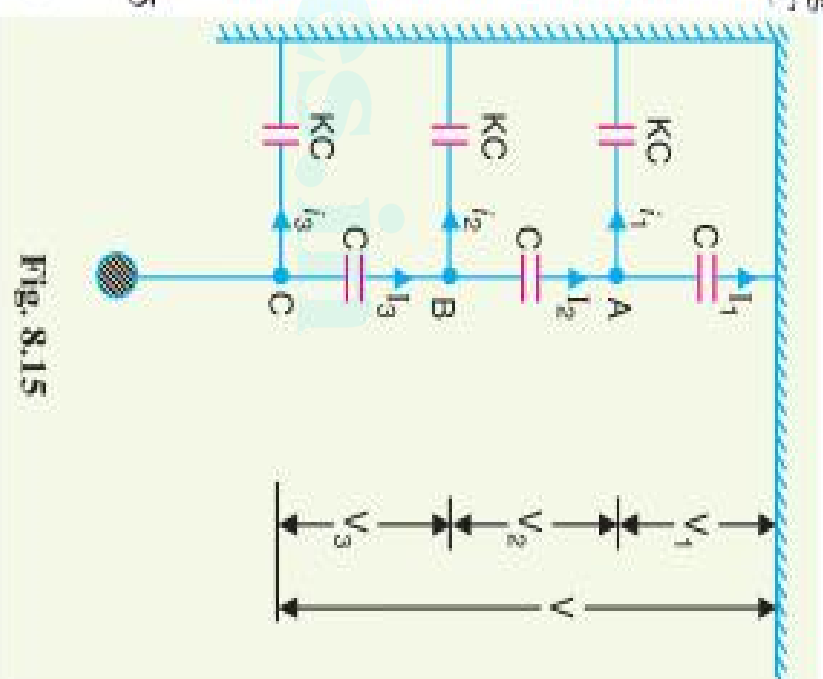


Fig. 8.15

Example 8.4. The three bus-bar conductors in an outdoor substation are supported by units of post type insulators. Each unit consists of a stack of 3 pin type insulators fixed one on the top of the other. The voltage across the lowest insulator is 13.1 kV and that across the next unit is 11 kV. Find the bus-bar voltage of the station.

Solution. The equivalent circuit of insulators is the same as shown in Fig. 8.15. It is given that $V_3 = 13.1$ kV and $V_2 = 11$ kV. Let K be the ratio of shunt capacitance to self capacitance of each unit. Applying Kirchhoff's current law to Junctions A and B, we can easily derive the following equations (See example 8.3) :

$$V_2 = V_1 (1 + K)$$

$$\text{or} \quad V_1 = \frac{V_2}{1 + K} \quad \dots (i)$$

$$\text{and} \quad V_3 = V_2 + (V_1 + V_2) K \quad \dots (ii)$$

Putting the value of V_1 = $V_2/(1 + K)$ in eq. (ii), we get,

$$V_3 = V_2 + \left[\frac{V_2}{1 + K} + V_2 \right] K$$

$$\text{or} \quad V_3 (1 + K) = V_2 (1 + K) + [V_2 + V_2 (1 + K)] K$$

$$= V_2 [(1 + K) + K + (K + K^2)]$$

$$= V_2 (1 + 3K + K^2)$$

$$\therefore 13.1 (1 + K) = 11 [1 + 3K + K^2]$$

$$\text{or} \quad 11K^2 + 19.9K - 2.1 = 0$$

Solving this equation, we get, $K = 0.1$.

$$\therefore V_1 = \frac{V_2}{1 + K} = \frac{11}{1 + 0.1} = 10 \text{ kV}$$

$$\text{Voltage between line and earth} = V_1 + V_2 + V_3 = 10 + 11 + 13.1 = 34.1 \text{ kV}$$

$$\therefore \text{Voltage between bus-bars (i.e., line voltage)}$$

Example 8.5. An insulator string consists of three units, each having a safe working voltage of 15 kV. The ratio of self-capacitance to shunt capacitance of each unit is 8 : 1. Find the maximum safe working voltage of the string. Also find the string efficiency.

Solution. The equivalent circuit of string insulators is the same as shown in Fig. 8.15. The maximum voltage will appear across the lowest unit in the string.

$$\therefore V_3 = 15 \text{ kV} ; \quad K = 1/8 = 0.125$$

Applying Kirchhoff's current law to junction A, we get,

$$V_2 = V_1 (1 + K)$$

$$\text{or} \quad V_1 = V_2 / (1 + K) = V_2 / (1 + 0.125) = 0.89 V_2$$

...(i)

Applying Kirchhoff's current law to Junction B, we get,

$$V_3 = V_2 + (V_1 + V_2) K = V_2 + (0.89 V_2 + V_2) \times 0.125$$

$\therefore$

$$V_3 = 1.236 V_2$$

...(ii)

$$\therefore \text{Voltage across middle unit, } V_2 = V_3 / 1.236 = 15 / 1.236 = 12.13 \text{ kV}$$

$$\text{Voltage across top unit, } V_1 = 0.89 V_2 = 0.89 \times 12.13 = 10.79 \text{ kV}$$

$$\text{Voltage across the String} = V_1 + V_2 + V_3 = 10.79 + 12.13 + 15 = 37.92 \text{ kV}$$

$$\begin{aligned} \text{String efficiency} &= \frac{37.92}{3 \times 15} \times 100 = 84.26 \% \end{aligned}$$

Example 8.6. A string of 4 insulators has a self-capacitance equal to 10 times the pin to earth capacitance. Find (i) the voltage distribution across various units expressed as a percentage of total voltage across the string and (ii) string efficiency.

Solution. When the number of insulators in a string exceeds 3, the nodal equation method becomes laborious. Under such circumstances, there is a simple method to solve the problem. In this method*, shunt capacitance (C_1) and self capacitance (C) of each insulator are represented by their equivalent reactances. As it is only the ratio of capacitances which determines the voltage distribution, therefore, the problem can be simplified by assigning unity value to X_C i.e., assuming $X_C = 1 \Omega$. If ratio of $C/C_1 = 10$, then we have $X_C = 1 \Omega$ and $X_{C1} = 10 \Omega$.

(i) Suppose $X_C = 1 \Omega$. As the ratio of self-capacitance to shunt capacitance (i.e., C/C_1) is 10, therefore, $X_{C1} = 10 \Omega$ as shown in Fig. 8.16 (i). Suppose that potential V across the string is such that 1 A current flows in the top insulator. Now the potential across each insulator can be easily determined. Thus :

Voltage across top unit, $V_1 = 1 \Omega \times 1 \text{ A} = 1 \text{ volt}$

Voltage across **2nd unit, $V_2 = 1 \Omega \times 1.1 \text{ A} = 1.1 \text{ volts}$

Voltage across †3rd unit, $V_3 = 1 \Omega \times 1.31 \text{ A} = 1.31 \text{ volts}$

Voltage across 4th unit, $V_4 = 1 \Omega \times 1.65 \text{ A} = 1.65 \text{ volts}$

Voltage obtained across the string $V = 1 + 1.1 + 1.31 + 1.65 = 5.06 \text{ volts}$

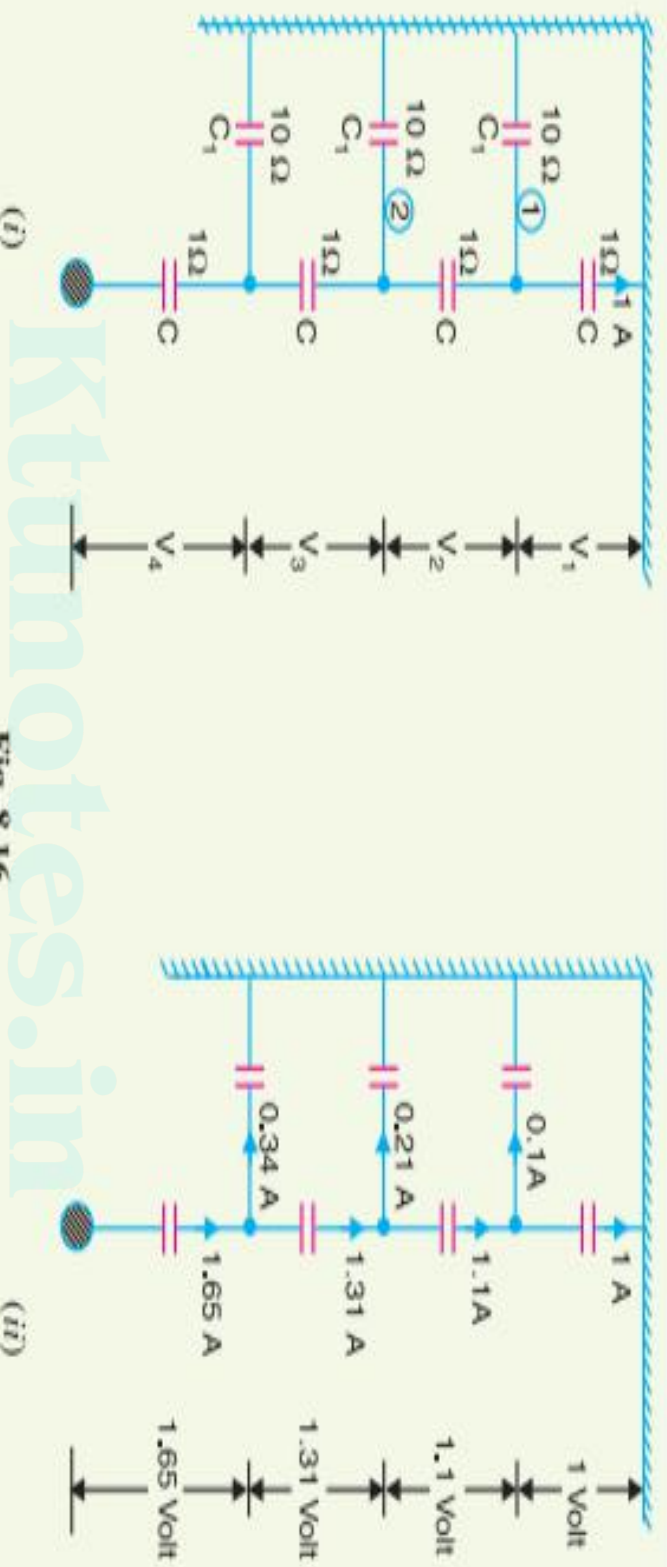


Fig. 8.16

The voltage across each unit expressed as a percentage of V (i.e., 5.06 volts) becomes :

Top unit

$$= (1/5.06) \times 100 = \mathbf{19.76\%}$$

Second from top

$$= (1.1/5.06) \times 100 = \mathbf{21.74\%}$$

Third from top

$$= (1.31/5.06) \times 100 = \mathbf{25.9\%}$$

Fourth from top

$$= (1.65/5.06) \times 100 = \mathbf{32.6\%}$$

(ii) String efficiency

$$= \frac{V}{4 \times V_4} \times 100 = \frac{5.06}{4 \times 1.65} \times 100 = \mathbf{76.6\%}$$

Corona

- When an alternating potential difference is applied across two conductors whose spacing is large, as compared to their diameters, there is no apparent change in the condition of atmospheric air surrounding the wires if the applied voltage is low.
- However, when the applied voltage exceeds a certain value, called *critical disruptive voltage*, the conductors are surrounded by a faint violet glow called corona.



- The phenomenon of corona is accompanied by a hissing sound, production of ozone, power loss and radio interference.
- The higher the voltage is raised, the larger and higher the luminous envelope becomes, and greater are the sound, the power loss and the radio noise.
- If the applied voltage is increased to breakdown value, a flash-over will occur between the conductors due to the breakdown of air insulation.
- *The phenomenon of violet glow, hissing noise and production of ozone gas in an overhead transmission line is known as **corona**.*

- If the conductors are polished and smooth, the corona glow will be uniform throughout the length of the conductors, otherwise the rough points will appear brighter.

- With d.c. voltage, there is difference in the appearance of the two wires.

- The positive wire has uniform glow about it, while the negative conductor has spotty glow

Theory of corona formation.

- Some ionisation is always present in air due to cosmic rays, ultra violet radiations and radioactivity.
- Therefore, under normal conditions, the air around the conductors contains some ionised particles (*i.e.*, free electrons and +ve ions) and neutral molecules.
- When p.d. is applied between the conductors, potential gradient is set up in the air which will have maximum value at the conductor surfaces. Under the influence of potential gradient, the existing free electrons acquire greater velocities.
- The greater the applied voltage, the greater the potential gradient and more is the velocity of free electrons.



- When the potential gradient at the conductor surface reaches about 30 kV per cm (max. value), the velocity acquired by the free electrons is sufficient to strike a neutral molecule with enough force to dislodge one or more electrons from it.
- This produces another ion and one or more free electrons, which in turn are accelerated until they collide with other neutral molecules, thus producing other ions.
- Thus, the process of ionisation is cumulative.
- The result of this ionisation is that either corona is formed or spark takes place between the conductors.

Factors Affecting Corona

- The phenomenon of corona is affected by the physical state of the atmosphere as well as by the conditions of the line. The following are the factors upon which corona depends :
- *Atmosphere.* As corona is formed due to ionisation of air surrounding the conductors, therefore, it is affected by the physical state of atmosphere. In the stormy weather, the number of ions is more than normal. Corona occurs at much less voltage as compared with fair weather.
- *Conductor size.* The corona effect depends upon the shape and conditions of the conductors. The rough and irregular surface will give rise to more corona because unevenness of the surface decreases the value of breakdown voltage. Thus a stranded conductor has irregular surface and hence gives rise to more corona than a solid conductor.

- *Spacing between conductors.* If the spacing between the conductors is made very large as compared to their diameters, there may not be any corona effect. It is because larger distance between conductors reduces the electro-static stresses at the conductor surface, thus avoiding corona formation.

- *Line voltage.* The line voltage greatly affects corona. If it is low, there is no change in the condition of air surrounding the conductors and hence no corona is formed. However, if the line voltage has such a value that electrostatic stresses developed at the conductor surface make the air around the conductor conducting, then corona is formed.

Important Terms

- The phenomenon of corona plays an important role in the design of an overhead transmission line. Therefore, it is profitable to consider the following terms much used in the analysis of corona effects:
- **Critical disruptive voltage.**
- *It is the minimum phase-neutral voltage at which corona occurs.*
- Consider two conductors of radii r cm and spaced d cm apart. If V is the phase-neutral potential, then potential gradient at the conductor surface is given by

$$g = \frac{V}{r \log_e \frac{d}{r}} \text{ volts / cm}$$

- In order that corona is formed, the value of g must be made equal to the breakdown strength of air. The breakdown strength of air at 76 cm pressure and temperature of 25°C is 30 kV/cm (*max*) or

21.2 kV/cm ($r.m.s.$) and is denoted by g_o . If V_c is the phase-neutral potential required under these conditions, then,

$$g_o = \frac{V_c}{r \log_e \frac{d}{r}}$$

where

g_o = breakdown strength of air at 76 cm of mercury and 25°C
 = 30 kV/cm (*max*) or 21.2 kV/cm (*r.m.s.*)

$$\therefore \text{Critical disruptive voltage, } V_c = g_o r \log_e \frac{d}{r}$$

The above expression for disruptive voltage is under standard conditions *i.e.*, at 76 cm of Hg and 25°C. However, if these conditions vary, the air density also changes, thus altering the value of g_o . The value of g_o is directly proportional to air density. Thus the breakdown strength of air at a barometric pressure of b cm of mercury and temperature of $t^\circ\text{C}$ becomes δg_o where

$$\delta = \text{air density factor} = \frac{3.92b}{273 + t}$$

Under standard conditions, the value of $\delta = 1$.

$$\therefore \text{Critical disruptive voltage, } V_c = g_o \delta r \log_e \frac{d}{r}$$

Correction must also be made for the surface condition of the conductor. This is accounted for by multiplying the above expression by irregularity factor m_o .

$$\therefore \text{Critical disruptive voltage, } V_c = m_o g_o \delta r \log_e \frac{d}{r} \text{ kV/phase}$$

where

$m_o = 1$ for polished conductors

= 0.98 to 0.92 for dirty conductors

deducted conductors

2. Visual critical voltage. *It is the minimum phase-neutral voltage at which corona glow appears all along the line conductors.*

- It has been seen that in case of parallel conductors, the corona glow does not begin at the disruptive voltage V_c but at a higher voltage V_v , called *visual critical voltage*. The phase-neutral effective value of visual critical voltage is given by the following empirical formula :

$$V_v = m_v g_o \delta r \left(1 + \frac{0.3}{\sqrt{\delta r}} \right) \log_e \frac{d}{r} \text{ kV/phase}$$

- where m_v is another irregularity factor having a value of 1.0 for polished conductors and 0.72 to 0.82 for rough conductors.

3. Power loss due to corona. Formation of corona is always accompanied by energy loss which is dissipated in the form of light, heat, sound and chemical action. When disruptive voltage is exceeded, the power loss due to corona is given by :

$$P = 242.2 \left(\frac{f+25}{\delta} \right) \sqrt{\frac{r}{d}} (V - V_c)^2 \times 10^{-5} \text{ kW / km / phase}$$

where

f = supply frequency in Hz

V = phase-neutral voltage (r.m.s.)

V_c = disruptive voltage (r.m.s.) per phase



Advantages and Disadvantages of Corona

Advantages

- Due to corona formation, the air surrounding the conductor becomes conducting and hence virtual diameter of the conductor is increased. The increased diameter reduces the electro-static stresses between the conductors.
- Corona reduces the effects of transients produced by surges.

Disadvantages

- Corona is accompanied by a loss of energy. This affects the transmission efficiency of the line.
- Ozone is produced by corona and may cause corrosion of the conductor due to chemical action.
- The current drawn by the line due to corona is non-sinusoidal and hence non-sinusoidal voltage drop occurs in the line. This may cause inductive interference with neighbouring communication lines

Methods of Reducing Corona Effect

- It has been seen that intense corona effects are observed at a working voltage of 33 kV or above. The corona effects can be reduced by the following methods :
- *By increasing conductor size.* By increasing conductor size, the voltage at which corona occurs is raised and hence corona effects are considerably reduced. This is one of the reasons that ACSR conductors which have a larger cross-sectional area are used in transmission lines.
- *By increasing conductor spacing.* By increasing the spacing between conductors, the voltage at which corona occurs is raised and hence corona effects can be eliminated. However, spacing cannot be increased too much otherwise the cost of supporting structure (e.g., bigger cross arms and supports) may increase to a considerable extent.
- **Example 8.13.** A 3-phase line has conductors 2 cm in diameter spaced equilaterally 1 m apart. If the dielectric strength of air is 30 kV (max) per cm, find the disruptive critical voltage for the line. Take air density factor $d = 0.952$ and irregularity factor $m = 0.9$.

Solution

Conductor radius,

$$r = 2/2 = 1 \text{ cm}$$

Conductor spacing,

$$d = 1 \text{ m} = 100 \text{ cm}$$

Dielectric strength of air, $g_o = 30 \text{ kV/cm (max.)} = 21.2 \text{ kV (r.m.s.) per cm}$

Disruptive critical voltage, $V_c = m_o g_o \delta r \log_e (d/r) \text{ kV}^*/\text{phase (r.m.s. value)}$

$$= 0.9 \times 21.2 \times 0.952 \times 1 \times \log_e 100/1 = 83.64 \text{ kV/phase}$$

$$\therefore \text{Line voltage (r.m.s.)} = \sqrt{3} \times 83.64 = \mathbf{144.8 \text{ kV}}$$

UNDER GROUND CABLES

An underground cable essentially consists of one or more conductors covered with suitable insulation and surrounded by a protecting cover. The conductor used in cables should be tinned stranded copper or aluminium of high conductivity. Stranding is done so that conductor may become flexible and carry more current.

- In general, a cable must fulfill the following necessary requirements :
 - The conductor used in cables should be tinned stranded copper or aluminium of high conductivity. Stranding is done so that conductor may become flexible and carry more current.
 - The conductor size should be such that the cable carries the desired load current without overheating and causes voltage drop within permissible limits.
 - The cable must have proper thickness of insulation in order to give high degree of safety and reliability at the voltage for which it is designed.
 - The cable must be provided with suitable mechanical protection so that it may withstand the rough use in laying it.
 - The materials used in the manufacture of cables should be such that there is complete chemical and physical stability throughout

CONSTRUCTION

The various parts are :

1. Cores or Conductors

A cable may have one or more than one core (conductor) depending upon the type of service for which it is intended. For instance, the 3-conductor cable shown in Fig. is used for 3-phase service.

The conductors are made of tinned copper or aluminium and are usually stranded in order to provide flexibility to the cable.

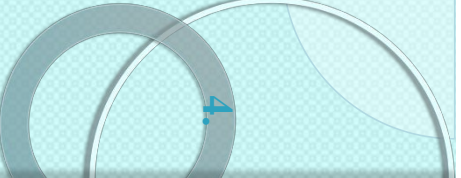
2. Insulation

Each core or conductor is provided with a suitable thickness of insulation, the thickness of layer depending upon the voltage to be withstood by the cable.

The commonly used materials for insulation are impregnated paper, varnished cambric or rubber mineral compound.

3. Metallic sheath

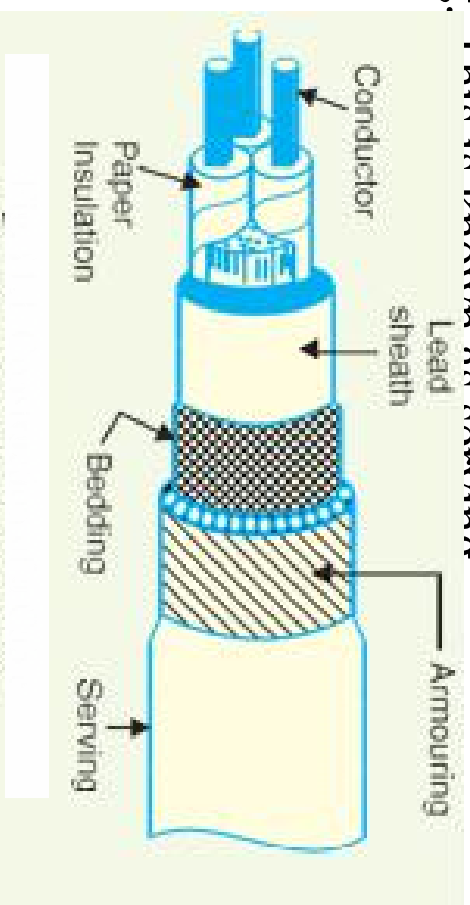
In order to protect the cable from moisture, gases or other damaging liquids (acids or alkalis) in the soil and atmosphere, a metallic sheath of lead or aluminium is provided over the insulation as shown in Fig.



Bedding. Over the metallic sheath is applied a layer of bedding which consists of a fibrous material like jute or hessian tape. The purpose of bedding is to protect the metallic sheath against corrosion and from mechanical injury due to armouring.

Armouring. Over the bedding, armouring is provided which consists of one or two layers of galvanised steel wire or steel tape. Its purpose is to protect the cable from mechanical injury while laying it and during the course of handling. Armouring may not be done in the case of some cables.

Serving. In order to protect armouring from atmospheric conditions, a layer of fibrous material (like jute) similar to bedding is provided over the armouring. This is known as serving.

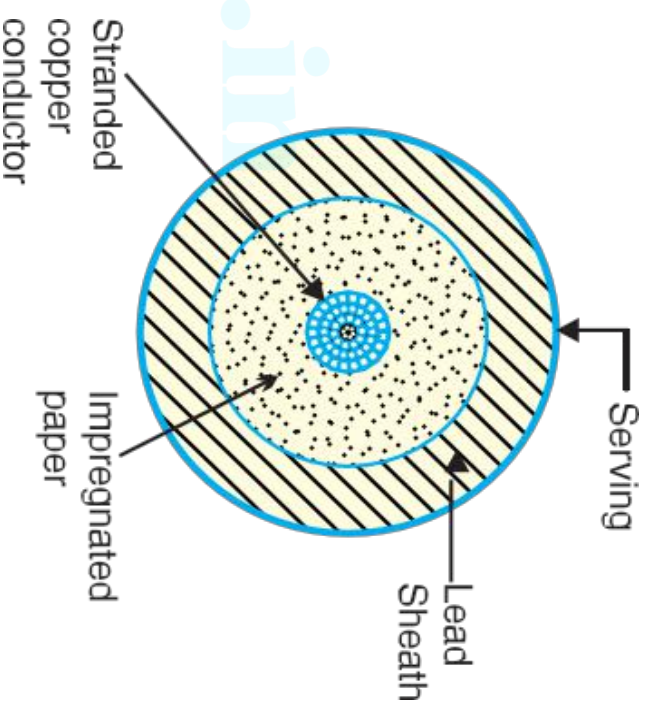


TYPES OF CABLES

- (i) the **type of insulating material** used in their manufacture
- (ii) the **voltage** for which they are manufactured.
 1. Low-tension (L.T.) cables — upto 1000 V
 2. High-tension (H.T.) cables — upto 11,000 V
 3. Super-tension (S.T.) cables — from 22 kV to 33 KV
 4. Extra high-tension (E.H.T.) cables — from 33 kV to 66 kV
 5. Extra super voltage cables — beyond 132 kV
- (iii) A cable may have **one or more than one core** depending upon the type of service for which it is intended. They are:
 - (i) single-core
 - (ii) two-core
 - (iii) three-core
 - (iv) four-core

Single-core low tension cable

- The cable has ordinary construction because the stresses developed in the cable for low voltages (upto 6600 V) are generally small.
- It consists of one circular core of tinned stranded copper (or aluminium) insulated by layers of impregnated paper.
- The insulation is surrounded by a lead sheath which prevents the entry of moisture into the inner parts.
- In order to protect the lead sheath from corrosion, an overall serving of compounded fibrous material (jute etc.) is provided.
- Single-core cables are not usually armoured in order to avoid excessive sheath losses.
- The principal advantages of single-core cables are simple construction and availability of larger copper section.



Three Core Cables

1. Belted cables — upto 11 kV
2. Screened cables — from 22 kV to 66 kV
3. Pressure cables — beyond 66 kV.

1. Belted cables. These cables are used for voltages upto 11kV

Fig. shows the constructional details of a 3-core belted cable.

The cores are insulated from each other by layers of impregnated paper. Another layer of impregnated paper tape, called *paper belt* is wound round the grouped insulated cores.

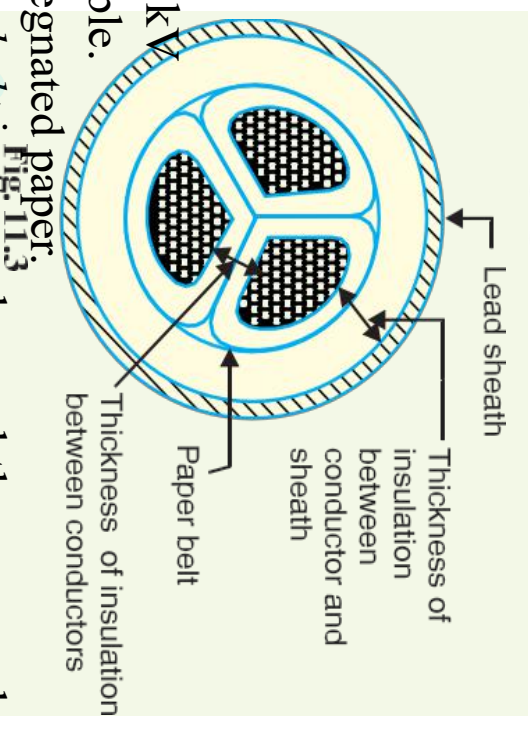
The gap between the insulated cores is filled with fibrous insulating material (jute etc.) so as to give circular cross-section to the cable.

The cores are generally stranded and may be of non- circular shape to make better use of available space.

The belt is covered with lead sheath to protect the cable against ingress of moisture and mechanical injury.

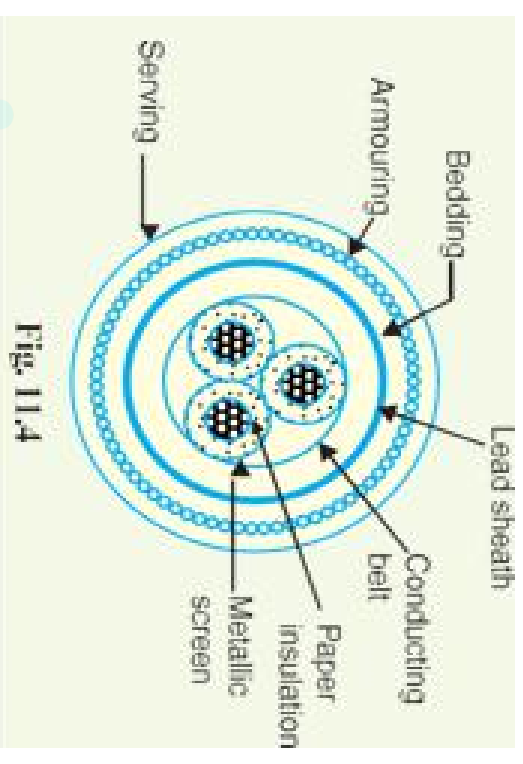
The lead sheath is covered with one or more layers of armouring with an outer serving.

The belted type construction is suitable only for low and medium voltages as the electrostatic stresses developed in the cables for these voltages are more or less radial *i.e.*, across the insulation.



H type

- This type of cable was first designed by H. Hochstadter and hence the name. Fig. shows the constructional details of a typical 3-core, *H*-type cable.
- Each core is insulated by layers of impregnated paper.
- The insulation on each core is covered with a metallic screen which usually consists of a perforated aluminium foil.
- The cores are laid in such a way that metallic screens make contact with one another.
- An additional conducting belt (copper woven fabric tape) is wrapped round the three cores.
- As all the four screens (3 core screens and one conducting belt) and the lead sheath are at earth potential, therefore, the electrical stresses are purely radial and consequently dielectric losses are reduced.
- Two principal advantages
 1. The perforations in the metallic screens assist in the complete impregnation of the cable with the compound and thus the possibility of air pockets or voids (vacuous spaces) in the dielectric is eliminated.
 2. The metallic screens



SL type cable

- It is basically *H*-type cable but the screen round each core insulation is covered by its own lead sheath.
- There is no overall lead sheath but only armouring and serving are provided.
- The S.L. type cables have two main **advantages** over *H*-type cables.
 1. Firstly, the separate sheaths minimise the possibility of core-to-core breakdown.
 2. Secondly, bending of cables becomes easy due to the elimination of overall lead sheath.
- The **disadvantage** is that the three lead sheaths of S.L. cable are much thinner than the single sheath of *H*-cable and, therefore, call for greater care in manufacture.

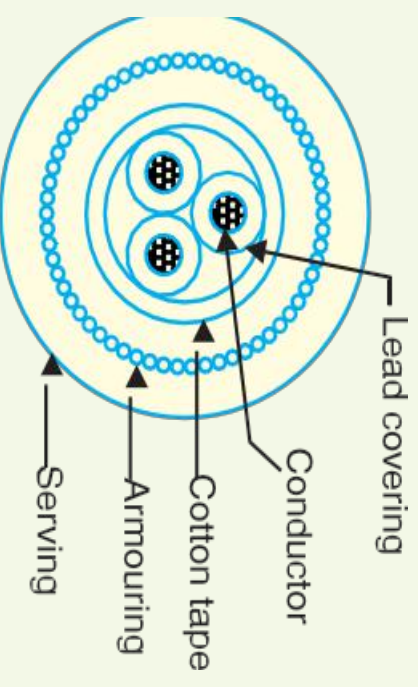


Fig. 11.5

Screened cables

- These cables are meant for use upto 33 kV, but in particular cases their use may be extended to operating voltages upto 66 kV.
- Two principal types of screened cables are H- type cables and S.L. type cables.
 - a) H-type cables*
 - b) S.L. type cables. (separate lead)*

3. Pressure cables

- For voltages beyond 66 kV, solid type cables are unreliable because there is a danger of breakdown of insulation due to the presence of voids.
- When the operating voltages are greater than 66 kV, *pressure cables* are used.
- In such cables, voids are eliminated by increasing the pressure of compound and for this reason they are called pressure cables.
- Two types of pressure cables
 - A. oil-filled cables
 - B. gas pressure cables are commonly used

a. Oil filled cables

- In such types of cables, channels or ducts are provided in the cable for oil circulation.
- The oil under pressure (it is the same oil used for impregnation) is kept constantly supplied to the channel by means of external reservoirs placed at suitable distances (say 500 m) along the route of the cable.
- Oil under pressure compresses the layers of paper insulation and is forced into any voids that may have formed between the layers.
- Due to the elimination of voids, oil-filled cables can be used for higher voltages, the range being from 66 kV upto 230 kV.
- Oil-filled cables are of three types viz., single-core conductor channel, single-core sheath channel and three-core filler-space channels.

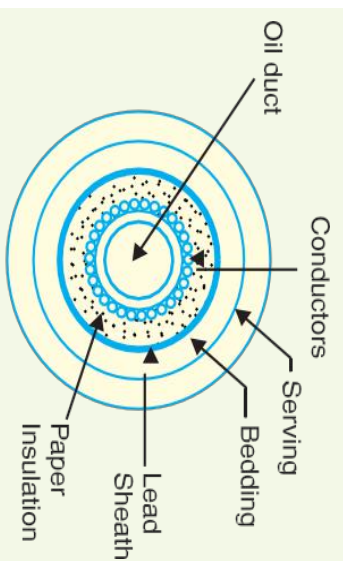


Fig. 11.6 Single-core conductor channel, oil-filled cable

B. Gas Pressure cable

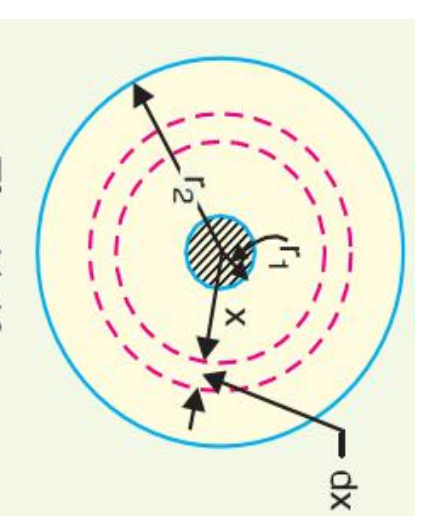
- The voltage required to set up ionisation inside a void increases as the pressure is increased.
- Therefore, if ordinary cable is subjected to a sufficiently high pressure, the ionisation can be altogether eliminated.
- At the same time, the increased pressure produces radial compression which tends to close any voids.
- This is the underlying principle of gas pressure cables.
- The triangular section reduces the weight and gives low thermal resistance but the main reason for triangular shape is that the lead sheath acts as a pressure membrane.
- Moreover, maintenance cost is small and the nitrogen gas helps in quenching any flame.
- However, it has the disadvantage that
- the overall cost is very high



Fig.

Insulation Resistance of a Single-Core Cable

- The cable conductor is provided with a suitable thickness of insulating material in order to prevent leakage current. The path for leakage current is radial through the insulation.
- **The opposition offered by insulation to leakage current is known as insulation resistance of the cable.**
- For satisfactory operation, the insulation resistance of the cable should be very high.
- Consider a single-core cable of conductor radius r_1 and internal sheath radius r_2 as shown in Fig. Let l be the length of the cable and ρ be the resistivity of the insulation.



- Consider a very small layer of insulation of thickness dx at a radius x . The length through which leakage current tends to flow is dx and the area of X-section offered to this flow is $2\pi x l$.

$\therefore$ Insulation resistance of considered layer

$$= \rho \frac{dx}{2\pi x l}$$

Insulation resistance of the whole cable is

$$R = \int_{r_1}^{r_2} \rho \frac{dx}{2\pi x l} = \frac{\rho}{2\pi l} \int_{r_1}^{r_2} \frac{1}{x} dx$$

$$\therefore R = \frac{\rho}{2\pi l} \log_e \frac{r_2}{r_1}$$

- **Problem** *A single-core cable has a conductor diameter of 1 cm and insulation thickness of 0.4 cm. If the specific resistance of insulation is $5 \times 10^{14} \Omega\text{cm}$, calculate the insulation resistance for a 2 km length of the cable.*

Ans: 234 M Ω

ktunotes.in

Solution

Conductor radius,

$$r_1 = 1/2 = 0.5 \text{ cm}$$

Length of cable,

$$l = 2 \text{ km} = 2000 \text{ m}$$

Resistivity of insulation,

$$\rho = 5 \times 10^{14} \Omega\text{-cm} = 5 \times 10^{12} \Omega\text{-m}$$

Internal sheath radius,

$$r_2 = 0.5 + 0.4 = 0.9 \text{ cm}$$

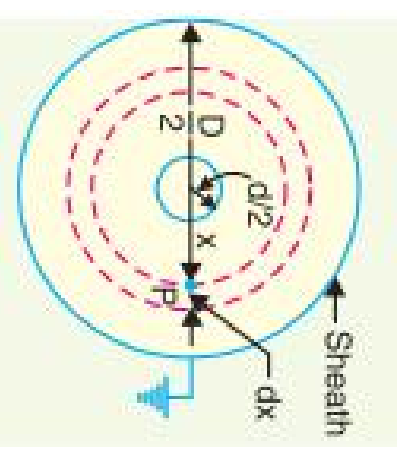
∴ Insulation resistance of cable is

$$\begin{aligned} R &= \frac{\rho}{2\pi l} \log_e \frac{r_2}{r_1} = \frac{5 \times 10^{12}}{2\pi \times 2000} \log_e \frac{0.9}{0.5} \\ &= 0.234 \times 10^9 \Omega = \mathbf{234 \text{ M}\Omega} \end{aligned}$$

Capacitance of Single core cable

- A single-core cable can be considered to be equivalent to two long co-axial cylinders.
- The conductor (or core) of the cable is the inner cylinder while the outer cylinder is represented by lead sheath which is at earth potential.
- Consider a single core cable with conductor diameter d and inner sheath diameter D (Fig).
- Let the charge per metre axial length of the cable be Q coulombs and ϵ be the permittivity of the insulation material between core and lead sheath.
- Obviously is relative permittivity.

$$\epsilon = \epsilon_0 \epsilon_r \text{ where } \epsilon_r$$



Consider a cylinder of radius x metres and axial length 1 metre. The surface area of this cylinder is $= 2 \pi x \times 1 = 2 \pi x \text{ m}^2$

$\therefore$ Electric flux density at any point P on the considered cylinder is

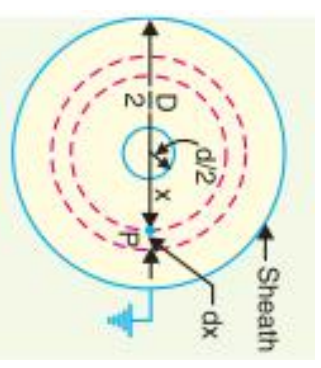
$$D_x = \frac{Q}{2 \pi x} \text{ C/m}^2$$

$$\text{Electric intensity at point } P, E_x = \frac{D_x}{\epsilon} = \frac{Q}{2 \pi x \epsilon} = \frac{Q}{2 \pi x \epsilon_0 \epsilon_r} \text{ volts/m}$$

- The work done in moving a unit positive charge from point P through a distance dx in the direction of electric field is $E_x dx$.

- Hence, the work done in moving a unit positive charge from conductor to sheath, which is the potential difference V between conductor and sheath, is given by :

$$V = \int_{d/2}^{D/2} E_x dx = \int_{d/2}^{D/2} \frac{Q}{2 \pi x \epsilon_0 \epsilon_r} dx = \frac{Q}{2 \pi \epsilon_0 \epsilon_r} \log_e \frac{D}{d}$$

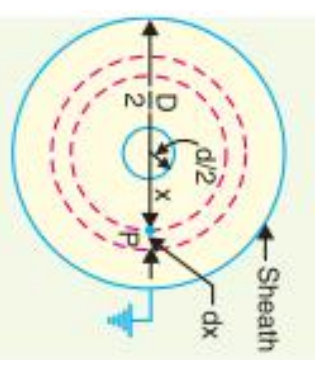


Capacitance of the cable is

$$\begin{aligned}
 C &= \frac{Q}{V} = \frac{Q}{\frac{Q}{2\pi\epsilon_0\epsilon_r} \log_{10} \frac{D}{d}} \text{ F/m} \\
 &= \frac{2\pi\epsilon_0\epsilon_r}{\log_{10}(D/d)} \text{ F/m} \\
 &= \frac{2\pi \times 8.854 \times 10^{-12} \times \epsilon_r}{2.303 \log_{10}(D/d)} \text{ F/m} \\
 &= \frac{41.4 \log_{10}(D/d)}{\epsilon_r} \times 10^{-9} \text{ F/m}
 \end{aligned}$$

If the cable has a length of l metres, then capacitance of the cable is

$$C = \frac{\epsilon_r l}{41.4 \log_{10} \frac{D}{d}} \times 10^{-9} \text{ F}$$



Example 11.5. Calculate the capacitance and charging current of a single core cable used on a 3-phase, 66 kV system. The cable is 1 km long having a core diameter of 10 cm and an impregnated paper insulation of thickness 7 cm. The relative permittivity of the insulation may be taken as 4 and the supply at 50 Hz.

Solution.

Capacitance of cable,
$$C = \frac{\epsilon_r l}{41.4 \log_{10}(D/d)} \times 10^{-9} \text{ F}$$

Here,

$$\epsilon_r = 4 ;$$

$$l = 1000 \text{ m}$$

$$d = 10 \text{ cm;}$$

$$D = 10 + 2 \times 7 = 24 \text{ cm}$$

Substituting these values in the above expression,

$$C = \frac{4 \times 1000}{41.4 \times \log_{10}(24/10)} \times 10^{-9} \text{ F} = 0.254 \times 10^{-6} \text{ F} = \mathbf{0.254 \mu F}$$

Voltage between core and sheath is

$$V_{ph} = 66/\sqrt{3} = 38.1 \text{ kV} = 38.1 \times 10^3 \text{ V}$$

$$\text{Charging current} = V_{ph}/X_C = 2\pi f C V_{ph}$$

$$= 2\pi \times 50 \times 0.254 \times 10^{-6} \times 38.1 \times 10^3 \text{ A} = \mathbf{3.04 \text{ A}}$$

Example 11.6. A 33 kV, 50 Hz, 3-phase underground cable, 4 km long uses three single core cables. Each of the conductor has a diameter of 2.5 cm and the radial thickness of insulation is 0.5 cm. Determine (i) capacitance of the cable/phase (ii) charging current/phase (iii) total charging kVAR. The relative permittivity of insulation is 3.

Solution.

$$(i) \text{ Capacitance of cable/phase, } C = \frac{\epsilon_r l}{41.4 \log_{10}(D/d)} \times 10^{-9} \text{ F}$$

$$\begin{array}{ll} \text{Here } \epsilon_r = 3 & ; \quad l = 4 \text{ km} = 4000 \text{ m} \\ d = 2.5 \text{ cm} & ; \quad D = 2.5 + 2 \times 0.5 = 3.5 \text{ cm} \end{array}$$

Putting these values in the above expression, we get,

$$C = \frac{3 \times 4000 \times 10^{-9}}{41.4 \times \log_{10}(3.5/2.5)} = 1984 \times 10^{-9} \text{ F}$$

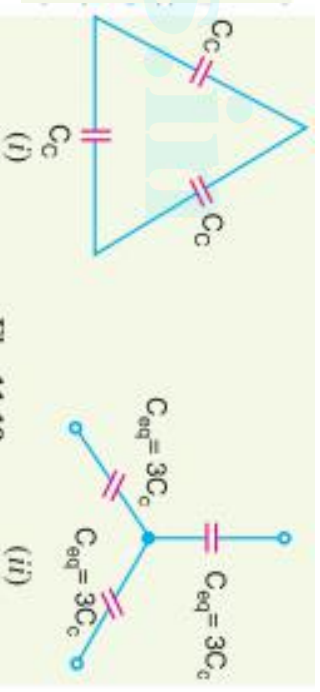
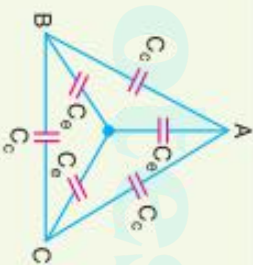
$$(ii) \text{ Voltage/phase, } V_{ph} = \frac{33 \times 10^3}{\sqrt{3}} = 19.05 \times 10^3 \text{ V}$$

$$\begin{aligned} \text{Charging current/phase, } I_C &= \frac{V_{ph}}{X_C} = 2\pi f C V_{ph} \\ &= 2\pi \times 50 \times 1984 \times 10^{-9} \times 19.05 \times 10^3 = 11.87 \text{ A} \end{aligned}$$

$$(iii) \text{ Total charging kVAR} = 3 V_{ph} I_C = 3 \times 19.05 \times 10^3 \times 11.87 = 678.5 \times 10^3 \text{ kVAR}$$

Capacitance of 3 CORE CABLE

- The capacitance of a cable system is much more important than that of overhead line because in cables (i) conductors are nearer to each other and to the earthed sheath (ii) they are separated by a dielectric of permittivity much greater than that of air.

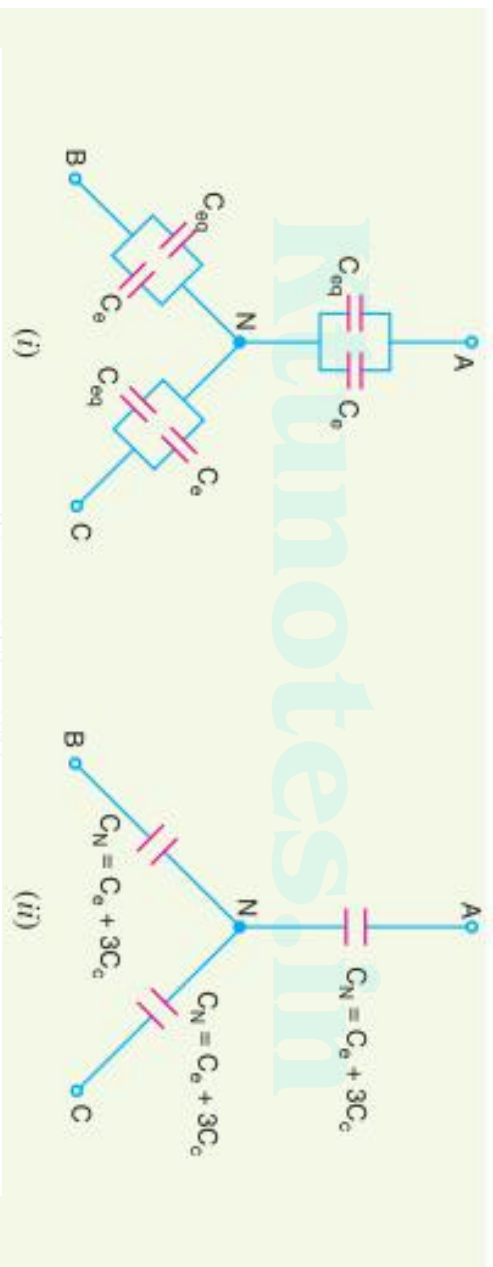


system.

- Since potential difference exists between pairs of conductors and between each conductor and the sheath, electrostatic fields are set up in the cable as shown in Fig.
- These electrostatic fields give rise to core-core capacitances C_c and conductor earth capacitances C_e as shown in Fig.

The three C_c are delta connected whereas the three C_e are star connected, the sheath forming the star point .

The three delta connected capacitances C_c [See Fig. 11.19 (i)] can be converted into equivalent star connected capacitances as shown in Fig. 11.19 (ii). It can be easily *shown that equivalent star- capacitance C_{eq} is equal to three times the delta- capacitance C_c i.e. $C_{eq} = 3C_c$.



$$C_N = C_e + C_{eq}$$

$$= C_e + 3C_c$$

If V_{ph} is the phase voltage, then charging current I_c is given by ;

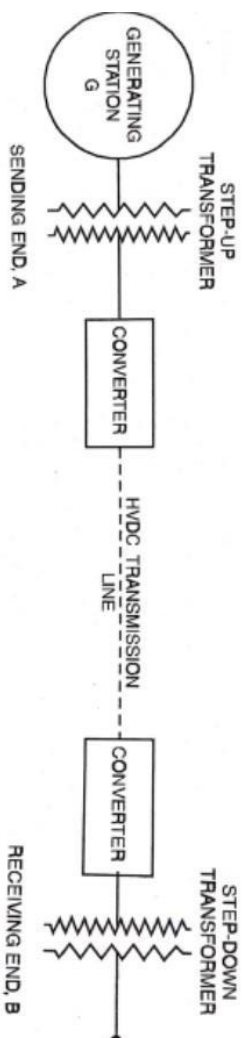
$$I_c = \frac{\text{Capacitive reactance per phase}}{V_{ph} C_N}$$

$$= \frac{2 \pi f V_{ph} C_N}{V_{ph} C_N}$$

HVDC – HIGH VOLTAGE DC TRANSMISSION

Principle of HVDC system operation

- When using direct current to provide an asynchronous link between 2 AC systems it is necessary to have to convert stations one at each end connected by a DC transmission line.
- The main equipments in a converter station are transformers and thyristor valves.
- Chokes and filters are provided at each end to ensure smooth direct current and suppress harmonics.
- At the sending and the thyristor valves act as rectifiers to convert AC into DC which is transmitted over the line.
- At the receiving end the thyristor valves act as inverters to convert DC into AC which is utilised at the receiving end.
- Single line diagram of hvdc transmission system is shown in figure



- A and B are the two converter stations.
- Converter station A is supplied from the generating station G.
- In converter station at the sending end the voltage is stepped up to appropriate value by step up transformer and then converted into direct current by thyristor valves.
- Thus at the start of transmission line we have high voltage DC .
- This rectified current flows along the transmission line to the receiving end converting station B where it is converted into three phase AC current by thyristor valves and then stepped down by the step down transformer to low voltage for further distribution.
- $P_S - P_R$ represents the power losses due to the conversion and transmission.

- The converter at the sending end acts as a rectifier and is suitable for power frequency (that is frequency of generator) on its AC side while converter at the receiving end acts as an inverter and its frequency is determined by frequency of the load system.
- This frequency is independent of the sending end frequency provided the two ends A and B are not additional connected by the three phase lines.
- The DC output voltage magnitude is controlled by varying the firing angle of the thyristor valves in the converter.
- In rectifier the firing angle is between zero degree and 90 degree while in inverter it is between 90 degree and 180 degree.
- As the DC output voltage is a function of cosine of the firing angle has the converter voltage becomes negative when firing angle α exceeds 90 degree.
- This makes the converter to operate as an inverter.
- The two converter at sending and receiving end are identical and whether they have to work as a rectifier or inverter is determined by the direction of power flow.
- In practical HVDC converter stations three phase bridge converters are employed at both ends. Reversible operation of converters as well as bidirectional power flow in hvdc link is possible simply by the control of firing angle..

Power control in HVDC system

- In DC system the power transferred from one station to another station is governed only by the magnitudes of terminal DC voltages at the two ends while in AC transmission system the power transfer is governed by phase difference (that is magnitude as well as phase) of the voltages at both ends.
- Thus the controllability of hvdc power is fast and stable.
- The current flows from higher voltage to lower voltage which are set by adjustment of firing or extinction angles of the two converters (that is rectifier and inverter.)
- If V_s is the voltage of the sending end and V_R we are is the voltage at the receiving end then the line current is given as
$$I_{dc} = \frac{V_s - V_R}{R}$$
- Where R is the resistance of the entire transmission link.
- The sending end voltage is given as
$$V_s = \left(\left(\frac{3\sqrt{2}}{\pi} V_{acs} \right) \cos \alpha - \frac{3X_{CS}}{\pi} I_{dc} \right)$$
- And receiving and voltage is given as
$$V_R = \left(\left(\frac{3\sqrt{2}}{\pi} V_{acR} \right) \cos \beta - \frac{3X_{CR}}{\pi} I_{dc} \right)$$
- here Alpha is the firing angle of rectifier beta is the extinction angle of inverter, V_{acs} is the AC side line to line RMS voltage at the sending end V_{acR} is the the AC side line to line RMS voltage at the receiving end .

- X_{cS} is the commutation reactance at the sending end
- X_{cR} is the computation reactance at the receiving end.
- Thus the power transferred is given as
- $P = V_S I_{dc} = \frac{V_S - V_R}{R} V_S$
- tap changes on the AC side take care of the voltage variations on AC side and DC power is controlled by controlling the sending and receiving and voltages V_S and V_R which is possible by control of firing and extinction angle alpha and beta respectively.

Table 14.1. Comparison of Long Distance Overhead EHV-AC and HDVC Transmission.

S.No.	Characteristics	EHV-AC System	HVDC System	Remarks
1.	Capital Cost. Line cost Substation Cost Number of circuits Intermediate Substations	Higher Lower More Required	Lower Higher One Not required	HVDC system becomes economical above 800 MW. Choice based on economics.
2.	Skin Effect	Present	Absent	Only single bipolar HVDC line adequate in most cases. HVDC substation has power losses. Single HVDC link adequate up to 3,000 MW
3.	Ground Return	Not possible	Possible	
4.	Line Losses	High	Low	
5.	Power Transfer	Limits imposed by power angle and line inductive reactance.	No limit	
6.	Voltage Control	Difficult for long lines due to series inductance and shunt capacitance	Easier	
7.	Compensation Requirements	Series and shunt compensations are necessary	No compensation is required	
8.	Corona and Radio Interference	More	Less	
9.	Reliability	Lesser	More	
10.	Short-circuit Currents	Large	Small	
11.	Control System	Simpler, cheaper and limitations of control	Difficult, costlier but fast and accurate control	For interconnections, HVDC system is advantageous due to superior control
12.	Stability of AC Networks	Very low due to line inductive reactance	Much higher	AC line is loaded up to 50% of its thermal rating while dc line is loaded up to 90% of its thermal rating

Advantages of HVDC system

1. Cheaper in cost

- Bipolar hvdc transmission lines require to pole conductors while AC transmission system requires three conductors to carry power.
- HVDC transmission can utilise Earth return and therefore does not require a double circuit while HVAC transmission always require a double circuit.
- The potential stress on the insulation in case of DC system is $\frac{1}{\sqrt{2}}$ times that of in AC system for the same operating voltage.
- As operating voltage less insulation is required.

2. HVDC line can be built in stages

- The DC line can be built as monopolar line with ground return in the initial stage and may be converted into a bipolar line on the later date when the load requirement increases.

3. No skin effect

- There is no skin effect in DC so there is a uniform distribution of current over the the section of the conductor that there is full utilisation of line conductor in case of DC transmission while it is not so in case of AC transmission.

4. Lower transmission losses

- HVDC transmission system needs only two conductors and therefore the power losses in a DC line are lesser than losses in DC line of the same power transfer capability.

5. Line loading

- The permissible loading on any HV AC line is limited by transient stability limit and line reactance to almost one third of thermal rating of conductors no such limit exist in case of hvdc lines.

6. greater reliability

7. lesser corona loss and radio interference

Limitations

1. **Costly terminal equipment**

- the converters required at both ends of the line have proved to be reliable but they are much more expensive than the conventional AC equipments.

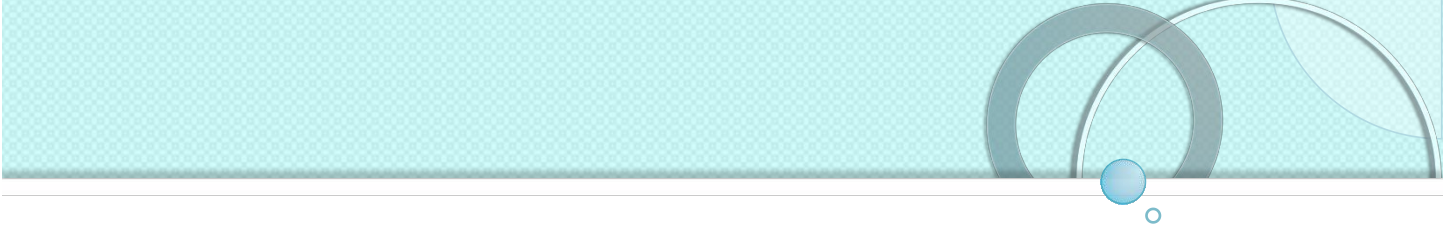
2. **More maintenance of line insulators**

- maintenance of insulators in hvdc transmission line is more

3. **Circuit breaking in multi-terminal DC systems is difficult and costly**

4. **Voltage transformation**

- voltage transformation is not easier in case of DC and hence it has to be accomplished on the AC side of the system



Thank You